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Dynamic Evacuation Plan Model for Aboagye Menyah Complex at KNUST

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A PROJECT SUBMITTED TO THE DEPARTMENT OF MATHEMATICS IN
PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF
BACHELOR OF SCIENCE MATHEMATICS

August 2023

Declaration

We hereby declare that this project is the result of our own and original research and that no part of it has been submitted to any institution or organization anywhere for the award of a degree. All inclusions from the work of others have been duly acknowledged

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Abstract

When a hazard, whether man-made or natural, occurs in buildings, areas or cities, there is the need to remove people, animals and belongings from those affected areas. Safely evacuating people from complex building structures during an emergency is a crucial problem for public safety. Using network optimization techniques, this study proposes an alternative method to address difficulty. The study focuses on the dynamic evacuation of a building complex, considering real-time considerations such as evolving danger situations. We begin by creating a detailed network model of the building complex, which includes spatial information, exit locations, and potential obstructions. We design adaptive evacuation techniques that minimize evacuation time while maximizing safety by utilizing advanced network optimization algorithms such as the Dijkstra algorithm. According to our findings, the dynamic evacuation method considerably improves the reliability and flexibility of complicated evacuation strategies. It allows evacuees and first responders to adapt quickly and effectively to changing emergencies, lowering risks and enhancing overall emergency readiness. The findings of this study contribute to the development of more robust and responsive evacuation procedures, which are critical for guaranteeing occupant safety and well-being in complex building environments.

Dedication

We dedicate this study to God.

Acknowledgements

We want to begin by expressing our heartfelt gratitude to the Almighty God for His blessing, protection, and guidance throughout our journey. We want to express our gratitude to Professor Peter Amoako-Yirenkyi, our supervisor, for his essential advice, constant support, and expertise throughout the research process. Your views and dedication have helped to shape the course of our work. Evander Nana Bosomefi Eghan, please accept our heartfelt thanks for your tireless efforts, unwavering counsel, and generous assistance. We are also grateful to our colleagues and fellow researchers who provided insightful input, contributed their knowledge, and engaged in insightful debates, all of which helped improve our research quality. Finally, we want to express our heartfelt gratitude to our friends and family for their constant encouragement and understanding throughout this endeavour's long hours and demands. This research was a collective effort, and the help and efforts of everyone involved were critical to its success. Thank you for joining us on this wonderful journey.

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Chapter 1

Introduction

1.1 Background of the Study

According to the definition provided by [Hamacher and Tjandra \(2001\)](#), the process of evacuating a building refers to the act of relocating individuals from a structure or a specific area inside it that has been identified as a hazardous zone to ensure safety as promptly as possible and with the utmost level of dependability. Each year, a significant number of deaths occur during emergency scenarios due to the lack of an optimal evacuation plan. According to the study conducted by [Rozo et al. \(2019\)](#), Consequently, some governments have instituted a need for municipal authorities to verify that evacuation preparations adhere to specific requirements [Darvishan and Lim \(2021\)](#). An emergency evacuation plan comprises a set of protocols designed to facilitate the process of evacuating individuals from a given location. A commonly implemented measure in building evacuation plans is providing occupants with instructions to locate and go towards the nearest exit point. The existing evacuation strategies frequently employ stationary illustrations or signage to indicate the closest exit or observation point. The study by [Cisek and Kapalka \(2014\)](#) highlights the potential for evacuating towards a hazard. According to research by [Lin et al. \(2008\)](#), most existing evacuation models presume that people will use predetermined optimal routes to their destinations and will escape most efficiently. To put it another way, optimisation models are rarely created. An optimisation model depicts inhabitants as a homogeneous flow circulating through an evacuation network, ignoring individual behaviour, and can be used as a benchmark to measure network efficiency if the network's inherent complexity is fully grasped. Traditionally, an optimisation model for evacuation problems is a static network model with constant features. As more

advanced optimisation approaches have become necessary, dynamic network flow models have piqued the interest of academics from numerous disciplines. The insufficiency of current evacuation plans in accurately representing the conditions during an emergency has been identified by [Rozo et al. \(2019\)](#). The primary objective of our research is to develop a dynamic evacuation plan for buildings, which incorporates real-time information obtained from the building during emergencies to adapt and modify evacuation routes.

1.2 Problem Statement

The primary issue identified in this study is the absence of dynamism in formulating evacuation plans. This deficiency derives from the reliance on pre-existing data about the building without considering the alterations induced by the emergency occurrence. This phenomenon renders The existing evacuation plans ineffective, posing a potential hazard.

1.3 Objectives

1. To design a static evacuation plan based on pre-existing knowledge of the building to facilitate safe and efficient leave procedures.
2. Develop a model for generating a dynamic plan for evacuating during an emergency using real-time information.
3. Conduct simulations utilising various scenarios to evaluate the efficacy of the suggested model.

1.4 Methodology

A network model was created for the case study building. This model was used to develop both static and dynamic optimisation models. These were solved using Dijkstra's

algorithm to find the shortest path to an exit from every intersection in the building.

1.5 Justification

This project aims to provide a method for determining the shortest alternate path to an exit in situations where the original evacuation path is obstructed. Including this feature in a dynamic evacuation system can facilitate the evacuation procedure during emergencies.

1.6 Case Study

The Aboagye Menyah Complex, located at Kwame Nkrumah University of Science and Technology, is our case study's focal point. The structure consists of numerous interconnected sections and pathways.

1.7 Organisation

This research consists of five primary sections. The initial chapter, the Introduction, encompasses many components, including the study's contextual backdrop, objectives, and problem statement. Chapter two of this research paper serves as a comprehensive literature review, encompassing a synthesis and analysis of pertinent studies conducted by various authors in the field. The third chapter of the research paper outlines the methods employed in the study, while the fourth chapter presents the findings and subsequent discussion. The concluding chapter of our study presents the final analysis and offers recommendations based on our findings.

Chapter 2

Literature Review

2.1 Introduction

Various kinds of research have been conducted, and articles have been written on evacuating people from a building, area or city using methods like the Bellman-Ford, A* and the Breadth-First Search algorithms. This study is focused on a dynamic evacuation plan for exiting building complexes using the Dijkstra algorithm.

2.1.1 Definitions

1. A building complex: This is a collection of buildings designed and constructed concurrently on a single site in a common architectural theme or for a certain use. These structures can work together as a single unit or separately and can be vertically or horizontally connected. Residential complexes, business complexes, or institutional complexes like campuses for businesses, hospitals, or schools can be classified as building complexes.
2. Evacuation: ([Hamacher and Tjandra \(2001\)](#)) defined evacuation as the organized and systematic process of removing people, animals, or assets from a potentially dangerous building or a part of it to a safer location. It is typically carried out in response to various emergencies and events, such as natural disasters (e.g., hurricanes, floods, wildfires), industrial accidents, chemical spills, terrorist threats, or other circumstances that pose a risk to human life, safety, or well-being.
3. Evacuation Plan: This is a detailed, predetermined strategy or set of procedures designed to safely and efficiently move people from a potentially dangerous or emer-

gency to a safer location. These plans are often developed for a variety of situations, including natural disasters (such as hurricanes, earthquakes, wildfires, floods, and tornadoes), industrial accidents, fires, terrorist threats, or any other occurrence that poses a risk to the safety of people in a specific area.

4. Graph Theory: Graph theory is a branch of mathematics that deals with the study of graphs. Graphs are mathematical structures used to model pairwise relationships between objects. In this study, the graph $G(V, E)$ consists of vertices(V), which are connected by edges(E). The graph in this study is weighted. Thus, the edges have a cost (distance).
5. Vertices(V): They are points connected by two or more lines. Vertices are also known as nodes, and in this study, they are the intersections of nodes in the building.
6. Edges(E): They are the lines that connect vertices. In this study, it represents the pathways and stairways in the building.
7. Weights(W): In this study, these are the distances on each edge.

2.2 Shortest Path Problem

The problem in this study is a shortest path problem. Shortest path problems (SPP) are a class of optimization problems in graph theory and network analysis that involve finding the shortest path or the minimum distance between two specific vertices (nodes) in a graph. The "shortest path" is typically defined as the path with the minimum sum of edge weights or costs. The shortest path problem is to find a route on a given graph between two vertices while reducing the total weights on its component's edges [Rohila et al. \(2014\)](#).

2.3 Shortest Path Algorithms

A shortest-path algorithm finds the shortest path between two vertices in a graph [Madkour et al. \(2017\)](#). Various algorithms are designed to solve shortest-path problems. Some of such algorithms are the Bellman-Ford algorithm, the Dijkstra algorithm and the Breadth-First Search algorithm.

2.3.1 Bellman-Ford Algorithm

The Bellman-Ford algorithm is a dynamic algorithm for finding the shortest paths in a weighted graph. It can be used for graphs with negative edge weights([Bannister and Eppstein \(2012\)](#)). It is a relaxation technique where an approximation to the correct distance is gradually replaced by more accurate values till an optimal solution is reached. It is slower than Dijkstra's algorithm for non-negative edge weights.

2.3.2 Dijkstra Algorithm

Dijkstra's algorithm is a well-known and frequently quoted algorithm for solving the single-source shortest paths for a given graph (V, E) with non-negative edge weights([Barbehenn \(1998\)](#)). The algorithm is primarily used for finding the shortest path from a single source node to all other nodes in a graph. The algorithm assumes that the graph is connected. The shortest path to the target node is discovered by iteratively expanding the shortest paths from a starting node to additional nodes in the graph.

2.3.3 Breadth-First Search Algorithm

Breadth-First Search (BFS) is an algorithm used to traverse or search through a graph or tree data structure by exploring all the neighbours of a node before moving to the next level of nodes([Rahim et al. \(2018\)](#)). BFS is typically used for finding the shortest path in an unweighted graph or for exploring the structure of a graph.

2.4 Literature Review

Many hazards threaten human populations, some of which are man-made and others that result from natural causes. The former include hazards such as chemical spills, accidents in nuclear power plants, and the collapse of large-scale construction works such as dam walls. Floods, earthquakes, and storms are examples of the latter. As society becomes reliant on technology, the risks from man-made causes appear to be on the rise. Indeed, one could argue that this increases our vulnerability to natural calamities as humanity squeezes ever more out of restricted earth ([Pidd et al. \(1996\)](#)).

Evacuation predictions are increasingly used in performance-based evaluations to evaluate the level of life safety given by buildings. In some circumstances, engineers utilize rough calculations to assess life safety, whereas evacuation models are used in others. Hand calculations often follow the formulae published in the Emergency Movement Chapter of the Society of Fire Protection Engineers (SFPE) Handbook to calculate mass flow evacuation from any height of a structure. As soon as the evacuation begins, the residents are envisioned to be standing at the entrance to the stairs on each level. The calculation focuses on areas of constriction within the building (often the door to the outside) and calculates the time it takes for occupants to flow past these points and to the outside. Engineers have been looking to evacuation computer models to assess a building's life safety for a more accurate evacuation calculation. There are several evacuation types to choose from, each with its own qualities and strengths ([Kuligowski et al. \(2005\)](#)).

Many individuals die in emergency circumstances every year due to a lack of effective evacuation strategies. The lack of a proper action plan indicates lengthy evacuation delays, which can endanger lives. People usually do not obey safety indications in dangerous situations; instead, they respond abruptly due to worry and anxiety to flee the area immediately. Most evacuation plans include a specific exit route and do not address the possibility of suggesting alternate routes in the event of obstructions or heavy traffic ([Rozo et al. \(2019\)](#)).

An article by (Wang et al. (2021)) also aimed to create the groundwork for a more realistic portrayal of human variables in evacuation models, which was required to assure the adequacy of infrastructure, decision processes, and evacuation safety. A generalized chronology was introduced to provide a clearer view of the empirical knowledge and modelling for evacuation investigations. Recent behavioural evidence from empirical research in both pedestrian and vehicular evacuations was examined to investigate the impact of various factors on evacuee behaviour at various stages. The emerging consensus viewpoint on essential behaviours was then utilized to analyze and consolidate recent achievements in evacuation modelling, particularly in formulations and approaches for capturing these behaviours.

Macroscopic models are primarily used to generate solid lower bounds for evacuation times and do not consider individual behaviour during an emergency. These boundaries can be used to examine existing buildings or aid in a new project's design phase. Macroscopic techniques based on dynamic network flow models are provided (least cost, maximum dynamic and universal maximum flow, fastest path and quickest flow). The described approach has the unique characteristic of allowing evacuee journey times to be density-dependent rather than constant. Priority regions and blockage due to fire or smoke may be assessed using multi-criteria optimization. It is demonstrated how to model utilizing temporal parameters, either discrete or continuous (Hamacher and Tjandra (2001))

The static evacuation plan depicts the building's static evacuation. This is intended to demonstrate the behaviour based on rescue and escape route planning, which is now widely employed in existing and new structures. One problem of static escape route direction is that it cannot respond to fluctuating events in the building. In the worst-case situation, the escape route straight leads into the fire or hazard zone. The dynamic evacuation plan depicts how an autonomous system could be used to evacuate an intelligent building. This indicates that the evacuation system is capable of reacting dynamically to threats (e.g., fire, toxic gases, or blocked corridors) and replanning escape routes in real-time and visualizing them to building residents (Wächter et al. (2021)).

Due to this, ([Wächter et al. \(2021\)](#)) experimented with static and dynamic evacuation plans. Both algorithms were tested using the following parameters: escape route blocked, evacuation time, and escape path length. The algorithms were compared for randomly generated scenarios and scenarios based on domain knowledge. According to the findings of this study, dynamic escape route guidance offers a high potential for facilitating building evacuation in an emergency event.

In their research, ([Van Willigen et al. \(2009\)](#)) offered WILLEM, a wireless sensor network-based system for dynamic evacuation routing in buildings. The method of dynamically calculating the fastest paths to exits is known as evacuation routing. If a fire breaks out elsewhere, the routes may be altered. They also offered an algorithm for detecting corridor congestion during evacuation and a method for providing persons in those congestions with an alternate path to the exit. Each phase of the system was thoroughly discussed, including the deployment of the wireless sensor network, the network's autonomous topology learning, and the real evacuation routing procedures. They created a simulation system that can mimic different types of evacuation pathways.

According to a paper by ([Shin et al. \(2019\)](#)), they described mathematical models and a heuristic approach for simultaneous evacuation and admission planning. Four mathematical models based on the discrete-time dynamic network flow model were created for simultaneous evacuation and entrance planning to provide the ideal pathways for evacuees and responders within a crucial time frame. The mathematical models' ideal paths could reduce the number of evacuees and responders in specified places. However, the mathematical model contained a flaw in the form of a long computation time for large-scale problems. They created a heuristic algorithm to bypass the constraint. They also conducted case studies to examine the characteristics of each model and the heuristic method. This research pioneered the field of evacuation planning by developing and analyzing four types of mathematical models and a heuristic algorithm which took into account simultaneous evacuation.

In a study by ([Cisek and Kapalka \(2014\)](#)), they concluded that the dynamic signage

system was a solution that considered both the hazard source and the distribution of individuals in buildings. Combining those two sets of information necessitated evacuation route assessment and enabled building evacuation optimization. The suggested model might optimize the evacuation process based on the assessment factors of choice, which included evacuation time, hazard proximity, and cumulative queuing time. The model also included an algorithm for determining all feasible evacuation strategies, represented as a series of graphs. This two-layer approach enabled quick calculations that could be completed in time for the results to be employed in an active dynamic evacuation signage system. The proposed method could be applied in buildings when there was a choice. An example of such a location was the junction of corridors from which it is possible to depart the building or a hallway on any floor of a high-rise building with at least two exits to two different staircases.

([Zhang and Chang \(2014\)](#)) focused their research on the dynamic evacuation model for pedestrian-vehicle mixed-flow networks. After their research, they concluded that their research offered an improved model for combining the optimization of mixed-flow movements and signal timings within the metropolitan area while accounting for conflicts between congested vehicle and pedestrian flows. The proposed model leveraged the shared node-link idea to represent the pedestrian and automobile networks, and connections were designed to reflect turning actions at crossings. The connecting node determined the flow conversion between pedestrian and vehicular flows. The proposed model, based on the locations of pick-up points for evacuees using transit systems and parking garage distributions for those with access to vehicles, was capable of producing a set of effective routing strategies to guide pedestrians and vehicles, as well as providing the optimal signal timings for either the dynamic or static signalling systems.

2.5 Summary

We learned from all of the articles we reviewed that dynamic evacuation strategies aided in the safe evacuation of evacuees. These publications provided numerous approaches and models. Following our evaluation, we employed the Dijkstra algorithm to create a dynamic plan for evacuating building complexes.

Chapter 3

Methodology

3.1 Introduction

This chapter outlines the process used to create a dynamic evacuation plan utilising network optimisation for a building complex. As part of our methodology, two optimisation models are formulated: one for “offline” settings and another for dynamic real-time scenarios. Both models are solved using Dijkstra’s algorithm. The main processes in our methodology are described in this chapter, starting with creating the network framework for the Aboagye Menyah Complex and ending with creating and solving the optimisation models.

3.2 Assumptions

To solve our problem, we made the following assumptions:

1. Evacuees are assumed to be at an intersection in the Aboagye Menyah complex at the start of an evacuation.
2. We consider macroscopic evacuation. We looked at the overall people flow at each evacuation path rather than individual evacuees and their peculiarities.

3.3 Model Description

3.3.1 Parameters

1. G = Undirected weight Graph.

$$G = (N, E, W)$$

2. N = Total collection of nodes (Intersections).
3. E = Total collection of edges connecting any two nodes.
4. W = Total collection of weights on each edge.
5. $x_{\alpha\beta}$ = Binary decision variable.

$$x_{\alpha\beta} = \begin{cases} 1, & \iff \text{there is a path between } \alpha \text{ and } \beta, \alpha\beta \in N. \\ 0, & \text{otherwise} \end{cases}$$

6. $F(\theta)$ = Total collection of edges leaving θ , $\theta \in N$
7. $R(\theta)$ = Total collection of edges entering θ , $\theta \in N$

3.3.2 Network Model

We developed a network model for the Aboagye Menyeh Complex. This is a weighted, undirected graph, $\mathbf{G}$. The building's stairways and paths make up the graph's edges, and the points where these pathways and stairways intersect are the nodes in the graph.

For the Abogye Menyeh Complex, we had a total of forty-three nodes. This comprised nine exit nodes, four basement nodes, seven ground-floor nodes, eight first-floor nodes, 8 second floor nodes and seven third-floor nodes. The representation of G is as follows:

$$\mathbf{G} = (N, E, W)$$

$$N = \{i \mid i \in [1, 43]\}$$

$$E = \{E_{\alpha\beta} \mid \alpha, \beta \in N\}$$

$$W = \{W_{\alpha\beta} \mid \alpha, \beta \in N\}$$

(3.1)

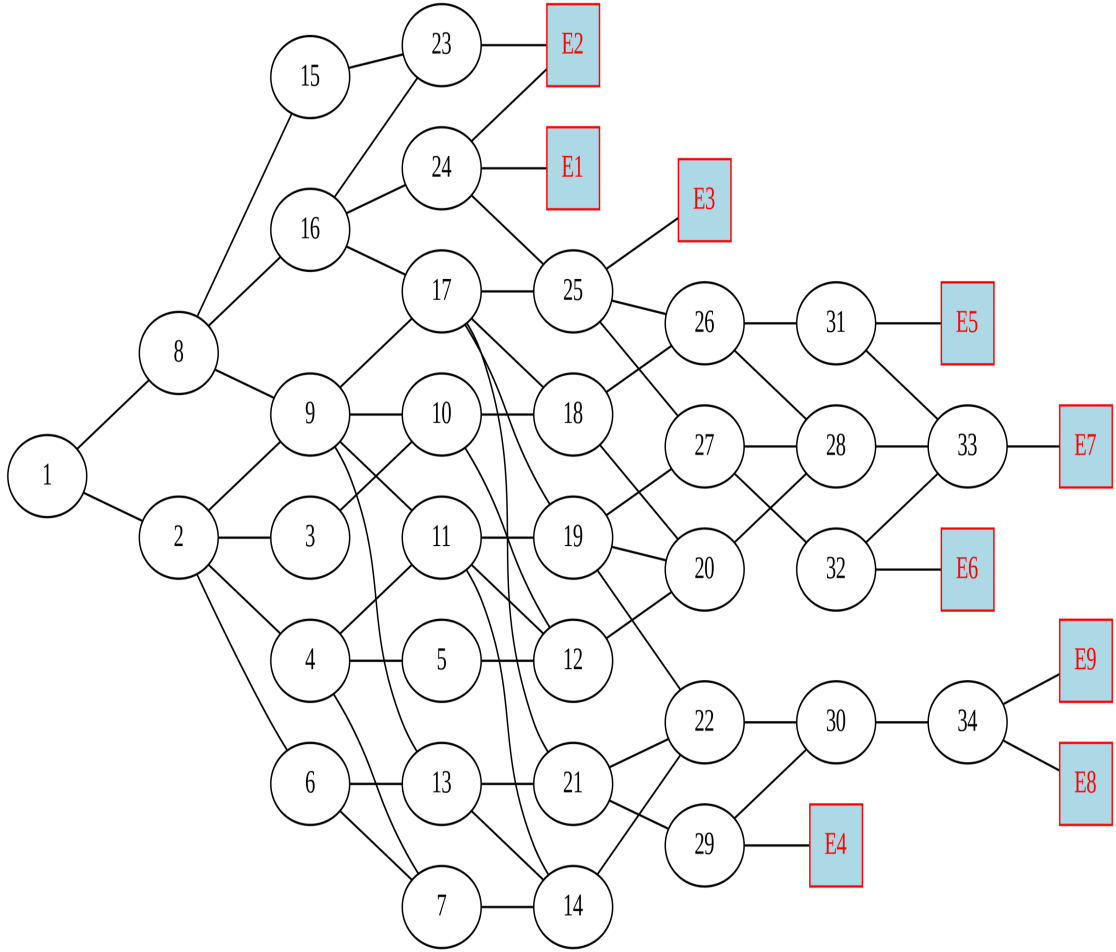


Figure 3.1: Graph $\mathbf{G}$: Network model of the Aboagye Menyah Complex

Actual distance of each pathway and stairway between adjacent nodes was measured in meters as the weight, $w_{\alpha\beta}$ on each edge, $e_{\alpha\beta}$. The weight values are shown in [Table 3.1](#).

Node Number	Weight(m)
1	$w_{1,2} = 22.8, w_{1,8} = 17.38$
2	$w_{2,3} = 21.87, w_{2,4} = 44.97, w_{2,6} = 39.16, w_{2,9} = 17.54$
3	$w_{3,10} = 17.25$
4	$w_{4,5} = 21.72, w_{4,7} = 40.14, w_{4,11} = 4.06$
5	$w_{5,12} = 17.25$
6	$w_{6,7} = 40.95, w_{6,13} = 23.68$
7	$w_{7,14} = 15.07$
8	$w_{8,9} = 22.8, w_{8,15} = 13.45, w_{8,16} = 17.38$
9	$w_{9,10} = 21.87, w_{9,11} = 44.97, w_{9,13} = 39.16, w_{9,17} = 17.54$
10	$w_{10,12} = 49.04, w_{10,18} = 17.25$
11	$w_{11,12} = 21.72, w_{11,14} = 40.14, w_{11,19} = 14.06$
12	$w_{12,20} = 17.25$
13	$w_{13,14} = 40.95, w_{13,21} = 23.68$
14	$w_{14,22} = 15.07$
15	$w_{15,23} = 11.38$
16	$w_{16,17} = 22.8, w_{16,23} = 13.45, w_{16,24} = 17.38$
17	$w_{17,18} = 21.87, w_{17,19} = 44.97, w_{17,21} = 39.16, w_{17,25} = 17.54$
18	$w_{18,20} = 49.04, w_{18,26} = 17.25$
19	$w_{19,20} = 21.72, w_{19,22} = 40.14, w_{19,27} = 14.06$
20	$w_{20,28} = 17.25$
21	$w_{21,22} = 40.95, w_{21,29} = 23.68$
22	$w_{22,30} = 15.07$
23	$w_{23,E2} = 11.38$
24	$w_{24,25} = 22.8, w_{24,E1} = 5.31, w_{24,E2} = 11.75$
25	$w_{25,26} = 21.87, w_{25,27} = 44.97, w_{25,E3} = 10.94$
26	$w_{26,28} = 49.04, w_{26,31} = 17.25$
27	$w_{27,28} = 21.72, w_{27,32} = 14.06$
28	$w_{28,33} = 17.25$
29	$w_{29,30} = 40.95, w_{29,E4} = 3.23$
30	$w_{30,34} = 15.07$
31	$w_{31,33} = 49.04, w_{31,E5} = 2.8$
32	$w_{32,33} = 24.6, w_{32,E6} = 8.14$
33	$w_{33,E7} = 13.89$
34	$w_{34,E8} = 11.35, w_{34,E9} = 10.6$

Table 3.1: Weights on each edge

3.3.3 Offline Optimisation

$$\min \sum_{\substack{\alpha\beta \in N \\ \alpha \neq \beta}} w_{\alpha\beta} x_{\alpha\beta} \quad (3.2)$$

subject to

$$\left(\sum_{e_{\alpha\beta} \in F(\theta)} x_{\alpha\beta} - \sum_{e_{\alpha\beta} \in R(\theta)} x_{\alpha\beta} \right) = \begin{cases} 1 & \text{if } \theta = s \\ -1 & \text{if } \theta = d \\ 0 & \text{Otherwise} \end{cases} \quad \forall \theta \in N \quad (3.3)$$

$$x_{\alpha\beta} \in \{0, 1\} \quad \forall \alpha\beta \in N \quad (3.4)$$

Equation 3.2 is the objective function for offline optimisation, which minimises the distance from a given source node, s , to a given destination node, d . Equation 3.3, a flow conservation constraint ensures the results form a continuous path which starts at s and ends at d . The trivial constraint, Equation 3.4 ensures the value of the decision variable $x_{\alpha\beta}$ is either zero or one.

3.3.4 Dynamic Optimisation

Considering V as a set of real-time blocked nodes, the real-time dynamic optimisation problem is given as:

$$\min \sum_{\substack{\alpha\beta \in N \\ \alpha\beta \notin V \\ \alpha \neq \beta}} w_{\alpha\beta} x_{\alpha\beta} \quad (3.5)$$

subject to

$$\left(\sum_{e_{\alpha\beta} \in F(\theta)} x_{\alpha\beta} - \sum_{e_{\alpha\beta} \in R(\theta)} x_{\alpha\beta} \right) = \begin{cases} 1 & \text{if } \theta = s \\ -1 & \text{if } \theta = d \\ 0 & \text{Otherwise} \end{cases} \quad \forall \theta \in N \quad (3.6)$$

$$x_{\alpha\beta} \in \{0, 1\} \quad \forall \alpha\beta \in N \quad (3.7)$$

[Equation 3.5](#) is the objective function for the dynamic optimisation, which minimises the distance from a given source node, s , to a given destination node, d , avoiding nodes that are in the set V . [Equation 3.6](#) and [Equation 3.7](#) are constraints, same as those in the offline optimisation.

3.4 Dijkstra's Algorithm

Dijkstra's method is well recognised for its utility in diverse domains, ranging from computer networking to GPS navigation systems, where the identification of the optimal route holds paramount importance. Dijkstra's method is renowned for its computational efficiency in addressing the single-source shortest path problem, offering a concise and best route to reach a given destination ([Skienna, 2009](#)). Dijkstra's algorithm generates a tree of shortest paths from a selected source node to all other nodes in the graph by iteratively searching nodes and updating distances. This is accomplished by guaranteeing that the path to each node that has been studied is the shortest one discovered thus far. Dijkstra's algorithm is pivotal for effectively traversing a network and prioritising efficiency ([Cormen et al., 2009](#)). It serves as a critical technique for resolving diverse and practical difficulties about routes and connections.

3.4.1 Algorithm

Algorithm 1 Dijkstra's Algorithm ([Even and Even, 2021](#))

```
1:  $\lambda(s) \leftarrow 0$ 
2:  $T \leftarrow \{s\}$ 
3:  $P \leftarrow \emptyset$ 
4: while  $T \neq \emptyset$  do
5:   Choose a vertex  $v \in T$  such that  $\lambda(v)$  is minimum
6:    $T \leftarrow T - \{v\}$ 
7:    $P \leftarrow T \cup \{v\}$ 
8:   for all  $v \rightarrow u$  do
9:     if  $u \in T$  then
10:       $\lambda(u) \leftarrow \min\{\lambda(u), \lambda(v) + w(v, u)\}$ 
11:     else if  $u \notin P$  then
12:       $\lambda(u) \leftarrow \lambda(v) + w(v, u)$ 
13:       $T \leftarrow T \cup \{u\}$ 
14:     end if
15:   end for
16: end while
```

The time complexity of Dijkstra's algorithm is dependent on how it is implemented. From this implementation, the algorithm has a run time of $O(|E| \cdot \log |V|)$

3.4.2 Procedure

The source node is designated as the initial node. The distance of node N refers to the measurement of the length between the start node and node N. The algorithm will iteratively assign weights to the nodes to improve their quality incrementally ([Ahuja et al., 1993](#)) .

1. Each node is assigned a preliminary distance value, with the source node assigned a value of zero and the alternative nodes assigned a value of N, where N represents a large numerical value.
2. The initial node must be referred to as "Present." Mark all alternate nodes as unvisited. Create a collection of all the nodes that have not yet been visited, referred to as the "unvisited set."

3. Considering the current node, it is necessary to consider all neighbouring nodes that have not yet been visited and calculate their initial distances. The user is advised to compare the newly calculated preliminary distance with the previously assigned number and select the lesser of the two.
4. Upon considering all the nodes directly connected to the current node, designate this node as visited and eliminate it from the collection of unvisited nodes. A previously visited vertex would not be revisited.
5. The process should be terminated once the last node has been marked as visited during the resolution of a route between two specific nodes or if the minimum preliminary distance among the set of unvisited nodes is equal to N during a complete traversal, indicating that there is no link connecting the starting node with the remaining unvisited nodes. Alternatively, select the unvisited node with the minimum preliminary distance and designate it as the "current node." Subsequently, go to the third step.

([Balakrishnan, 1995](#))

Chapter 4

Results and Discussing

4.1 Introduction

In this chapter, we show the results of our work to use network optimisation to make a dynamic escape plan for a building complex. Using the method described in the last chapter, we show the data, analyses, and results in tables and graphs that show the effectiveness of our approach.

4.2 Results of Offline Optimisation

The offline optimisation model employed Dijkstra's algorithm to construct a static evacuation plan for the Aboagye Menyah Complex. The computation results are presented in [Table 4.1](#). The provided data displays the minimum distance between each of the thirty-four (34) non-exit nodes and each of the nine (9) exit nodes, as well as the cumulative distance of these paths.

Node	Exit	Shortest Path	Distance(m)
1	E1	1-8-16-24-E1	57.37
1	E2	1-8-15-23-E2	53.51
1	E3	1-8-16-24-25-E3	85.8
1	E4	1-8-16-17-21-29-E4	123.55

1	E5	1-2-3-10-18-26-31-E5	162.71
1	E6	1-2-4-11-19-27-32-E6	132.15
1	E7	1-2-3-10-18-26-31-33-E7	130.36
1	E8	1-2-6-7-14-22-30-34-E8	171.54
1	E9	1-2-6-7-14-22-30-34-E9	170.79
2	E1	2-1-8-16-24-E1	80.17
2	E2	2-1-8-15-23-E2	76.31
2	E3	2-9-17-25-E3	63.56
2	E4	2-9-17-21-29-E4	101.15
2	E5	2-3-10-18-26-31-E5	139.91
2	E6	2-4-11-19-27-32-E6	109.35
2	E7	2-3-10-18-26-31-33-E7	107.56
2	E8	2-6-7-14-22-30-34-E8	148.74
2	E9	2-6-7-14-22-30-34-E9	147.99
3	E1	3-10-18-26-25-24-E1	101.73
3	E2	3-10-9-8-15-23-E2	98.13
3	E3	3-10-18-26-25-E3	84.56
3	E4	3-10-18-17-21-29-E4	122.44

3	E5	3-10-18-26-31-E5	118.04
3	E6	3-10-18-26-31-33-32-E6	104.54
3	E7	3-10-18-26-31-33-E7	85.69
3	E8	3-2-6-7-14-22-30-34-E8	170.61
3	E9	3-2-6-7-14-22-30-34-E9	169.86
4	E1	4-11-19-27-25-24-E1	115.26
4	E2	4-11-9-8-15-23-E2	118.04
4	E3	4-11-19-27-25-E3	98.09
4	E4	4-11-19-22-30-29-E4	127.51
4	E5	4-11-19-27-32-33-31-E5	132.68
4	E6	4-11-19-27-32-E6	64.38
4	E7	4-11-19-27-32-33-E7	94.73
4	E8	4-11-19-22-30-34-E8	109.75
4	E9	4-11-19-22-30-34-E9	109.0
5	E1	5-4-11-19-27-25-24-E1	136.98
5	E2	5-4-11-9-8-15-23-E2	139.76
5	E3	5-4-11-19-27-25-E3	119.81
5	E4	5-4-11-19-22-30-29-E4	149.23

5	E5	5-12-20-28-33-31-E5	120.84
5	E6	5-4-11-19-27-32-E6	86.1
5	E7	5-12-20-28-33-E7	82.89
5	E8	5-4-11-19-22-30-34-E8	131.47
5	E9	5-4-11-19-22-30-34-E9	130.72
6	E1	6-2-1-8-16-24-E1	116.33
6	E2	6-2-1-8-15-23-E2	112.47
6	E3	6-2-9-17-25-E3	99.72
6	E4	6-13-21-29-E4	74.27
6	E5	6-2-3-10-18-26-31-E5	176.07
6	E6	6-7-4-11-19-27-32-E6	145.47
6	E7	6-2-3-10-18-26-31-33-E7	143.72
6	E8	6-7-14-22-30-34-E8	112.58
6	E9	6-7-14-22-30-34-E9	111.83
7	E1	7-4-11-19-27-25-24-E1	155.4
7	E2	7-6-2-1-8-15-23-E2	152.92
7	E3	7-4-11-19-27-25-E3	138.23
7	E4	7-14-22-30-29-E4	89.39

7	E5	7-4-11-19-27-32-33-31-E5	172.82
7	E6	7-4-11-19-27-32-E6	104.52
7	E7	7-4-11-19-27-32-33-E7	134.87
7	E8	7-14-22-30-34-E8	71.63
7	E9	7-14-22-30-34-E9	70.88
8	E1	8-16-24-E1	40.07
8	E2	8-15-23-E2	36.21
8	E3	8-16-24-25-E3	68.5
8	E4	8-16-17-21-29-E4	106.25
8	E5	8-9-10-18-26-31-E5	145.46
8	E6	8-9-11-19-27-32-E6	118.09
8	E7	8-9-10-18-26-31-33-E7	113.11
8	E8	8-9-13-14-22-30-34-E8	159.47
8	E9	8-9-13-14-22-30-34-E9	158.72
9	E1	9-8-16-24-E1	62.87
9	E2	9-8-15-23-E2	59.01
9	E3	9-17-25-E3	46.02
9	E4	9-17-21-29-E4	83.61

9	E5	9-10-18-26-31-E5	122.66
9	E6	9-11-19-27-32-E6	95.29
9	E7	9-10-18-26-31-33-E7	90.31
9	E8	9-13-14-22-30-34-E8	136.67
9	E9	9-13-14-22-30-34-E9	135.92
10	E1	10-18-26-25-24-E1	84.48
10	E2	10-9-8-15-23-E2	80.88
10	E3	10-18-26-25-E3	67.31
10	E4	10-18-17-21-29-E4	105.19
10	E5	10-18-26-31-E5	100.79
10	E6	10-18-26-31-33-32-E6	87.29
10	E7	10-18-26-31-33-E7	68.44
10	E8	10-9-13-14-22-30-34-E8	158.54
10	E9	10-9-13-14-22-30-34-E9	157.79
11	E1	11-19-27-25-24-E1	101.2
11	E2	11-9-8-15-23-E2	103.98
11	E3	11-19-27-25-E3	84.03
11	E4	11-19-22-30-29-E4	113.45

11	E5	11-19-27-32-33-31-E5	118.62
11	E6	11-19-27-32-E6	50.32
11	E7	11-19-27-32-33-E7	80.67
11	E8	11-19-22-30-34-E8	95.69
11	E9	11-19-22-30-34-E9	94.94
12	E1	12-20-28-33-31-26-25-24-E1	121.78
12	E2	12-11-9-8-15-23-E2	125.7
12	E3	12-20-28-33-31-26-25-E3	104.61
12	E4	12-11-19-22-30-29-E4	135.17
12	E5	12-20-28-33-31-E5	103.59
12	E6	12-11-19-27-32-E6	72.04
12	E7	12-20-28-33-E7	65.64
12	E8	12-11-19-22-30-34-E8	117.41
12	E9	12-11-19-22-30-34-E9	116.66
13	E1	13-9-8-16-24-E1	102.03
13	E2	13-9-8-15-23-E2	98.17
13	E3	13-9-17-25-E3	85.18
13	E4	13-21-29-E4	50.59

13	E5	13-9-10-18-26-31-E5	161.82
13	E6	13-14-11-19-27-32-E6	131.41
13	E7	13-9-10-18-26-31-33-E7	129.47
13	E8	13-14-22-30-34-E8	97.51
13	E9	13-14-22-30-34-E9	96.76
14	E1	14-22-21-17-16-24-E1	140.67
14	E2	14-13-9-8-15-23-E2	139.12
14	E3	14-22-21-17-25-E3	123.66
14	E4	14-22-30-29-E4	74.32
14	E5	14-11-19-27-32-33-31-E5	158.76
14	E6	14-11-19-27-32-E6	90.46
14	E7	14-11-19-27-32-33-E7	120.81
14	E8	14-22-30-34-E8	56.56
14	E9	14-22-30-34-E9	55.81
15	E1	15-23-E2-24-E1	39.82
15	E2	15-23-E2	22.76
15	E3	15-23-E2-24-25-E3	68.25
15	E4	15-23-16-17-21-29-E4	113.7

15	E5	15-23-E2-24-25-26-31-E5	145.47
15	E6	15-23-E2-24-25-27-32-E6	124.48
15	E7	15-23-E2-24-25-26-31-33-E7	113.12
15	E8	15-23-16-17-21-22-30-34-E8	169.23
15	E9	15-23-16-17-21-22-30-34-E9	168.48
16	E1	16-24-E1	22.69
16	E2	16-23-E2	24.83
16	E3	16-24-25-E3	51.12
16	E4	16-17-21-29-E4	88.87
16	E5	16-17-18-26-31-E5	128.21
16	E6	16-17-19-27-32-E6	104.03
16	E7	16-17-18-26-31-33-E7	95.86
16	E8	16-17-21-22-30-34-E8	144.4
16	E9	16-17-21-22-30-34-E9	143.65
17	E1	17-16-24-E1	45.49
17	E2	17-16-23-E2	47.63
17	E3	17-25-E3	28.48
17	E4	17-21-29-E4	66.07

17	E5	17-18-26-31-E5	105.41
17	E6	17-19-27-32-E6	81.23
17	E7	17-18-26-31-33-E7	73.06
17	E8	17-21-22-30-34-E8	121.6
17	E9	17-21-22-30-34-E9	120.85
18	E1	18-26-25-24-E1	67.23
18	E2	18-17-16-23-E2	69.5
18	E3	18-26-25-E3	50.06
18	E4	18-17-21-29-E4	87.94
18	E5	18-26-31-E5	83.54
18	E6	18-26-31-33-32-E6	70.04
18	E7	18-26-31-33-E7	51.19
18	E8	18-17-21-22-30-34-E8	143.47
18	E9	18-17-21-22-30-34-E9	142.72
19	E1	19-27-25-24-E1	87.14
19	E2	19-17-16-23-E2	92.6
19	E3	19-27-25-E3	69.97
19	E4	19-22-30-29-E4	99.39

19	E5	19-27-32-33-31-E5	104.56
19	E6	19-27-32-E6	36.26
19	E7	19-27-32-33-E7	66.61
19	E8	19-22-30-34-E8	81.63
19	E9	19-22-30-34-E9	80.88
20	E1	20-28-33-31-26-25-24-E1	104.53
20	E2	20-28-33-31-26-25-24-E2	110.97
20	E3	20-28-33-31-26-25-E3	87.36
20	E4	20-19-22-30-29-E4	121.11
20	E5	20-28-33-31-E5	86.34
20	E6	20-19-27-32-E6	57.98
20	E7	20-28-33-E7	48.39
20	E8	20-19-22-30-34-E8	103.35
20	E9	20-19-22-30-34-E9	102.6
21	E1	21-17-16-24-E1	84.65
21	E2	21-17-16-23-E2	86.79
21	E3	21-17-25-E3	67.64
21	E4	21-29-E4	26.91

21	E5	21-17-18-26-31-E5	144.57
21	E6	21-22-19-27-32-E6	117.35
21	E7	21-17-18-26-31-33-E7	112.22
21	E8	21-22-30-34-E8	82.44
21	E9	21-22-30-34-E9	81.69
22	E1	22-21-17-16-24-E1	125.6
22	E2	22-21-17-16-23-E2	127.74
22	E3	22-21-17-25-E3	108.59
22	E4	22-30-29-E4	59.25
22	E5	22-19-27-32-33-31-E5	144.7
22	E6	22-19-27-32-E6	76.4
22	E7	22-19-27-32-33-E7	106.75
22	E8	22-30-34-E8	41.49
22	E9	22-30-34-E9	40.74
23	E1	23-E2-24-E1	28.44
23	E2	23-E2	11.38
23	E3	23-E2-24-25-E3	56.87
23	E4	23-16-17-21-29-E4	102.32

23	E5	23-E2-24-25-26-31-E5	134.09
23	E6	23-E2-24-25-27-32-E6	113.1
23	E7	23-E2-24-25-26-31-33-E7	101.74
23	E8	23-16-17-21-22-30-34-E8	157.85
23	E9	23-16-17-21-22-30-34-E9	157.1
24	E1	24-E1	5.31
24	E2	24-E2	11.75
24	E3	24-25-E3	33.74
24	E4	24-16-17-21-29-E4	106.25
24	E5	24-25-26-31-E5	110.96
24	E6	24-25-27-32-E6	89.97
24	E7	24-25-26-31-33-E7	78.61
24	E8	24-16-17-21-22-30-34-E8	161.78
24	E9	24-16-17-21-22-30-34-E9	161.03
25	E1	25-24-E1	28.11
25	E2	25-24-E2	34.55
25	E3	25-E3	10.94
25	E4	25-17-21-29-E4	83.61

25	E5	25-26-31-E5	88.16
25	E6	25-27-32-E6	67.17
25	E7	25-26-31-33-E7	55.81
25	E8	25-17-21-22-30-34-E8	139.14
25	E9	25-17-21-22-30-34-E9	138.39
26	E1	26-25-24-E1	49.98
26	E2	26-25-24-E2	56.42
26	E3	26-25-E3	32.81
26	E4	26-18-17-21-29-E4	105.19
26	E5	26-31-E5	66.29
26	E6	26-31-33-32-E6	52.79
26	E7	26-31-33-E7	33.94
26	E8	26-31-33-32-27-19-22-30-34-E8	154.4
26	E9	26-31-33-32-27-19-22-30-34-E9	153.65
27	E1	27-25-24-E1	73.08
27	E2	27-25-24-E2	79.52
27	E3	27-25-E3	55.91
27	E4	27-19-22-30-29-E4	113.45

27	E5	27-32-33-31-E5	90.5
27	E6	27-32-E6	22.2
27	E7	27-32-33-E7	52.55
27	E8	27-19-22-30-34-E8	95.69
27	E9	27-19-22-30-34-E9	94.94
28	E1	28-33-31-26-25-24-E1	87.28
28	E2	28-33-31-26-25-24-E2	93.72
28	E3	28-33-31-26-25-E3	70.11
28	E4	28-27-19-22-30-29-E4	135.17
28	E5	28-33-31-E5	69.09
28	E6	28-27-32-E6	43.92
28	E7	28-33-E7	31.14
28	E8	28-27-19-22-30-34-E8	117.41
28	E9	28-27-19-22-30-34-E9	116.66
29	E1	29-21-17-16-24-E1	108.33
29	E2	29-21-17-16-23-E2	110.47
29	E3	29-21-17-25-E3	91.32
29	E4	29-E4	3.23

29	E5	29-21-17-18-26-31-E5	168.25
29	E6	29-30-22-19-27-32-E6	132.42
29	E7	29-21-17-18-26-31-33-E7	135.9
29	E8	29-30-34-E8	67.37
29	E9	29-30-34-E9	66.62
30	E1	30-22-21-17-16-24-E1	140.67
30	E2	30-22-21-17-16-23-E2	142.81
30	E3	30-22-21-17-25-E3	123.66
30	E4	30-29-E4	44.18
30	E5	30-22-19-27-32-33-31-E5	159.77
30	E6	30-22-19-27-32-E6	91.47
30	E7	30-22-19-27-32-33-E7	121.82
30	E8	30-34-E8	26.42
30	E9	30-34-E9	25.67
31	E1	31-26-25-24-E1	67.23
31	E2	31-26-25-24-E2	73.67
31	E3	31-26-25-E3	50.06
31	E4	31-26-18-17-21-29-E4	122.44

31	E5	31-E5	49.04
31	E6	31-33-32-E6	35.54
31	E7	31-33-E7	16.69
31	E8	31-33-32-27-19-22-30-34-E8	137.15
31	E9	31-33-32-27-19-22-30-34-E9	136.4
32	E1	32-27-25-24-E1	87.14
32	E2	32-27-25-24-E2	93.58
32	E3	32-27-25-E3	69.97
32	E4	32-27-19-22-30-29-E4	127.51
32	E5	32-33-31-E5	76.44
32	E6	32-E6	8.14
32	E7	32-33-E7	38.49
32	E8	32-27-19-22-30-34-E8	109.75
32	E9	32-27-19-22-30-34-E9	109.0
33	E1	33-31-26-25-24-E1	70.03
33	E2	33-31-26-25-24-E2	76.47
33	E3	33-31-26-25-E3	52.86
33	E4	33-31-26-18-17-21-29-E4	125.24

33	E5	33-31-E5	51.84
33	E6	33-32-E6	32.74
33	E7	33-E7	13.89
33	E8	33-32-27-19-22-30-34-E8	134.35
33	E9	33-32-27-19-22-30-34-E9	133.6
34	E1	34-30-22-21-17-16-24-E1	155.74
34	E2	34-30-22-21-17-16-23-E2	157.88
34	E3	34-30-22-21-17-25-E3	138.73
34	E4	34-30-29-E4	59.25
34	E5	34-30-22-19-27-32-33-31-E5	174.84
34	E6	34-30-22-19-27-32-E6	106.54
34	E7	34-30-22-19-27-32-33-E7	136.89
34	E8	34-E8	11.35
34	E9	34-E9	10.6

Table 4.1: Results of Offline Optimisation

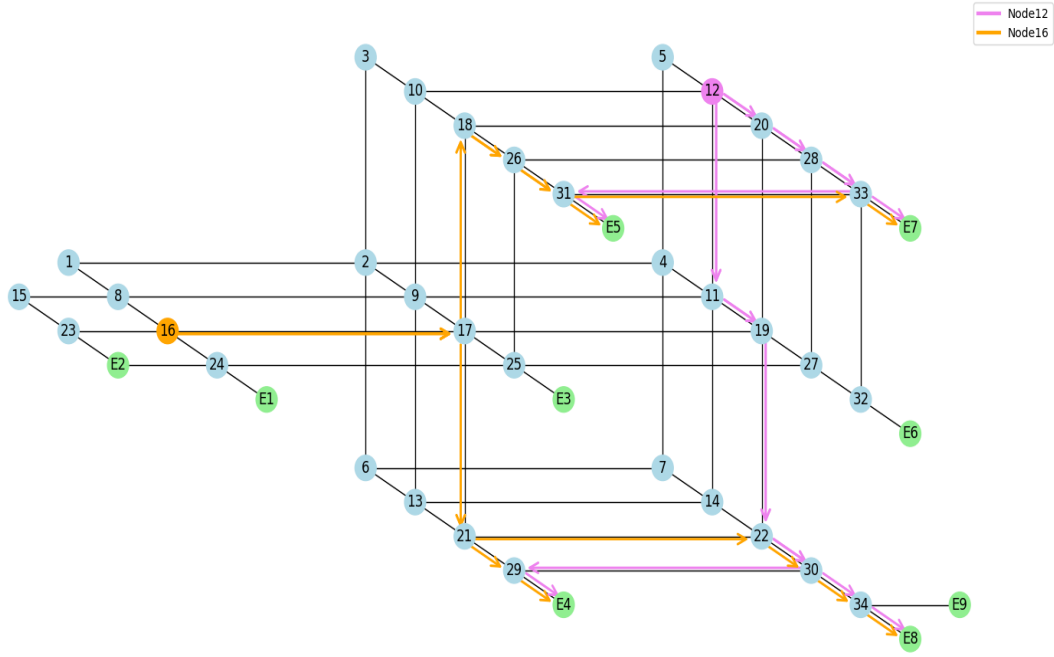


Figure 4.1: Shortest path from nodes 12 and 16 to different exits

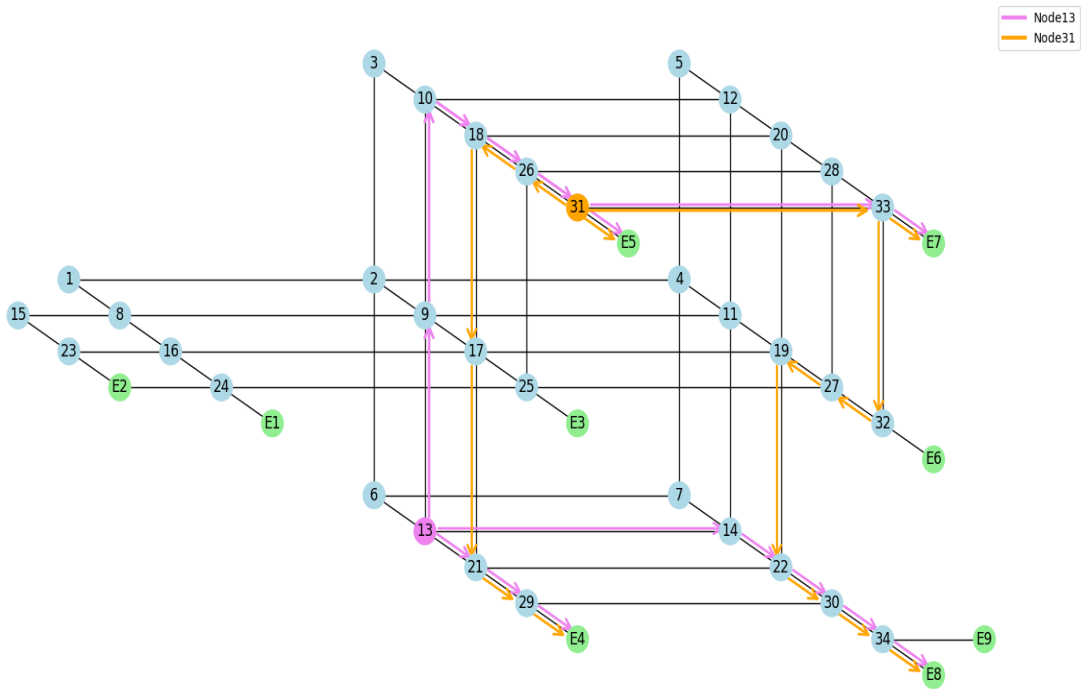


Figure 4.2: Shortest path from nodes 13 and 31 to different exits

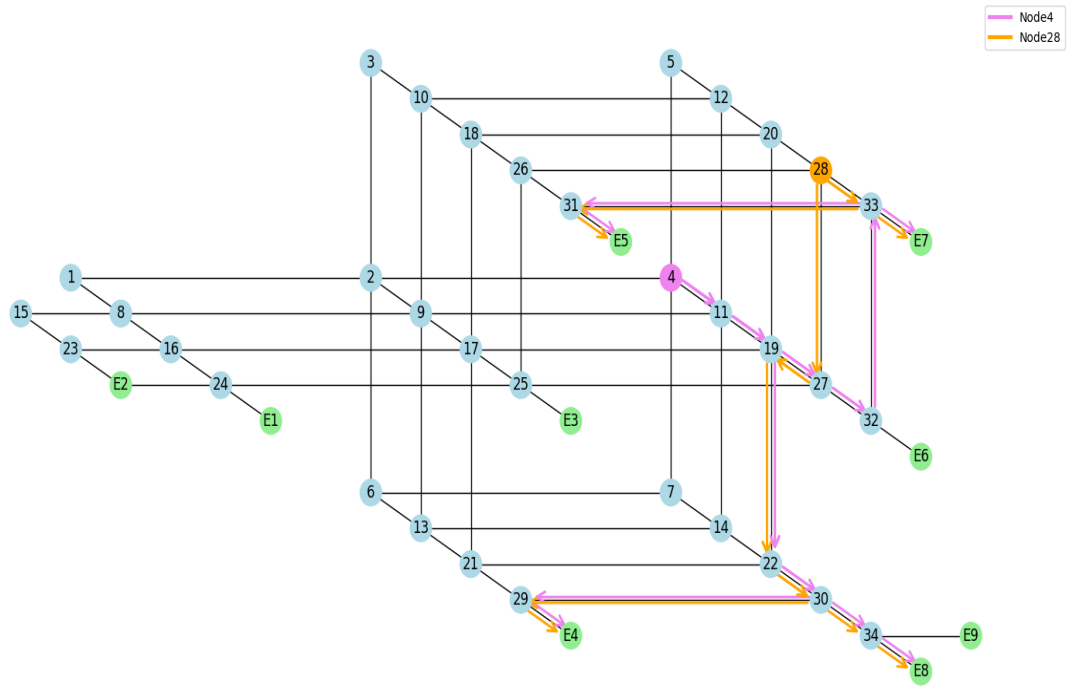


Figure 4.3: Shortest path from nodes 4 and 28 to different exits

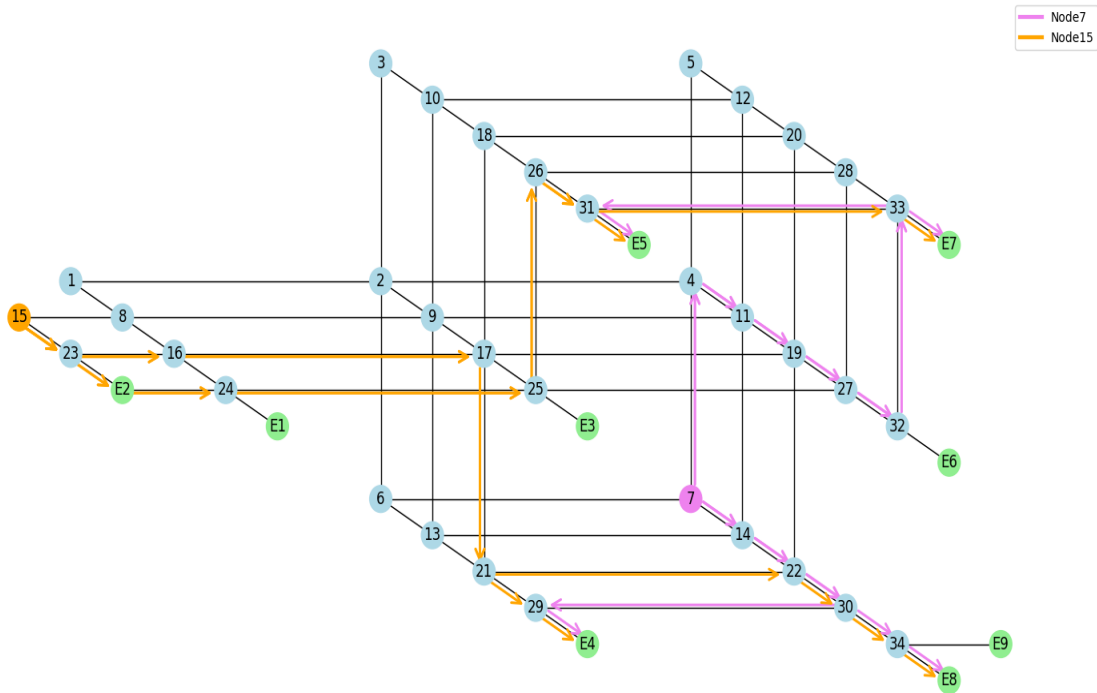


Figure 4.4: Shortest path from nodes 7 and 15 to different exits

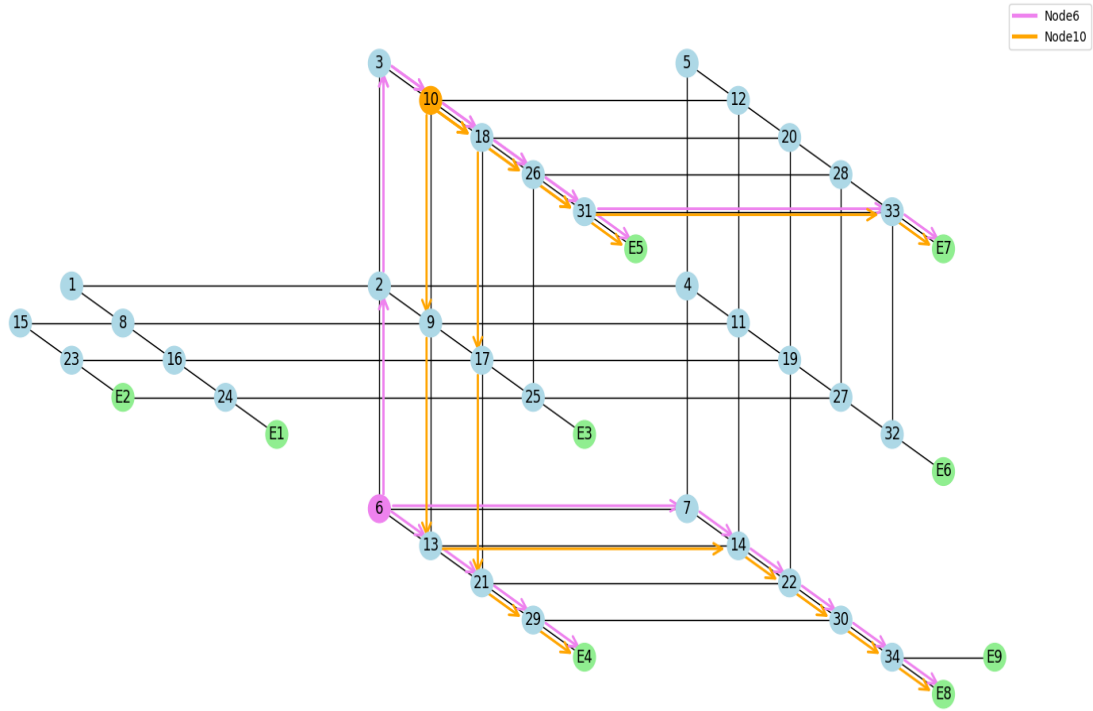


Figure 4.5: Shortest path from nodes 6 and 10 to different exits

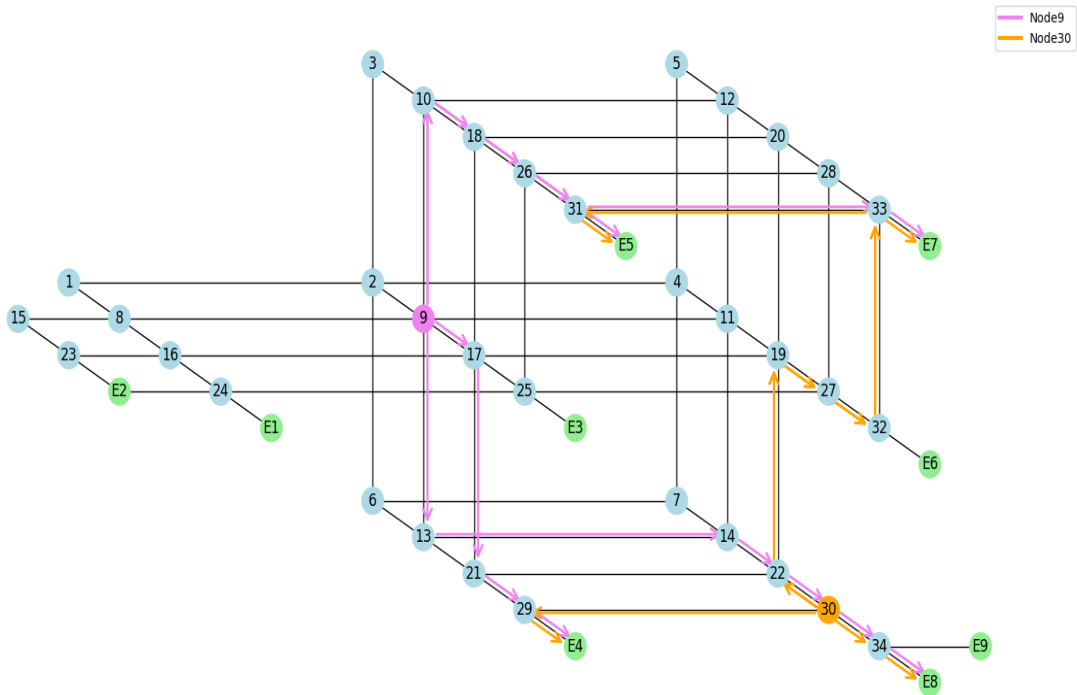


Figure 4.6: Shortest path from nodes 9 and 30 to different exits

The figures presented above illustrate a subset of the results that were obtained. The graphs provided depict the structural depiction of the Aboagye Menyeh Complex network model. The illustrated pathways depict the shortest routes from a given node to the exits E4, E5, E7, and E8, as determined through the implementation of Dijkstra's algorithm. **Figure 4.1** shows the shortest paths from nodes 12 and 16 to selected edges. **Figure 4.2** shows the shortest paths from nodes 13 and 31 to selected nodes. **Figure 4.3**, **Figure 4.4**, **Figure 4.5**, **Figure 4.6** illustrates the optimum path to selected exits for nodes 4 and 28, 7 and 15, 6 and 10 and 9 and 30 respectfully.

4.3 Results for Dynamic Optimisation

To evaluate the efficacy of our dynamic model, we conducted multiple scenarios involving randomly selected blocked nodes.

4.3.1 Scenario A

Consider the set $V = \{6, 19, 12, 17, 26\}$, representing a collection of real-time blocked nodes. Upon using Dijkstra's algorithm to the dynamic optimisation model, any route originating from the stationary evacuation plan that consists of any of the nodes in V will be circumvented. An alternative route was produced that excludes the obstructed nodes. **Table 4.2** displays the alternate route determined by the dynamic model for each route that contains a blocked node in its offline evacuation route.

Node	Exit	Shortest Path	Distance(m)
1	E4	1-8-9-13-21-29-E4	129.85
1	E5	1-8-16-24-25-27-32-33-31-E5	210.33
1	E6	1-8-16-24-25-27-32-E6	142.03

1	E7	1-8-16-24-25-27-32-33-E7	172.38
1	E8	1-8-9-13-14-22-30-34-E8	176.77
1	E9	1-8-9-13-14-22-30-34-E9	176.02
2	E3	2-1-8-16-24-25-E3	108.6
2	E4	2-9-13-21-29-E4	107.29
2	E5	2-3-10-18-20-28-33-31-E5	191.75
2	E6	2-1-8-16-24-25-27-32-E6	164.83
2	E7	2-3-10-18-20-28-33-E7	153.8
2	E8	2-9-13-14-22-30-34-E8	154.21
2	E9	2-9-13-14-22-30-34-E9	153.46
3	E1	3-10-9-8-16-24-E1	101.99
3	E3	3-10-9-8-16-24-25-E3	130.42
3	E4	3-10-9-13-21-29-E4	128.87
3	E5	3-10-18-20-28-33-31-E5	169.88
3	E6	3-10-18-20-28-27-32-E6	144.71
3	E7	3-10-18-20-28-33-E7	131.93
3	E8	3-10-9-13-14-22-30-34-E8	175.79
3	E9	3-10-9-13-14-22-30-34-E9	175.04

4	E1	4-11-9-8-16-24-E1	121.9
4	E3	4-11-9-8-16-24-25-E3	150.33
4	E4	4-11-14-22-30-29-E4	128.52
4	E5	4-11-9-10-18-20-28-33-31-E5	233.53
4	E6	4-11-9-8-16-24-25-27-32-E6	206.56
4	E7	4-11-9-10-18-20-28-33-E7	195.58
4	E8	4-11-14-22-30-34-E8	110.76
4	E9	4-11-14-22-30-34-E9	110.01
5	E1	5-4-11-9-8-16-24-E1	143.62
5	E3	5-4-11-9-8-16-24-25-E3	172.05
5	E4	5-4-11-14-22-30-29-E4	150.24
5	E5	5-4-11-9-10-18-20-28-33-31-E5	255.25
5	E6	5-4-11-9-8-16-24-25-27-32-E6	228.28
5	E7	5-4-11-9-10-18-20-28-33-E7	217.3
5	E8	5-4-11-14-22-30-34-E8	132.48
5	E9	5-4-11-14-22-30-34-E9	131.73
7	E1	7-14-13-9-8-16-24-E1	158.05
7	E2	7-14-13-9-8-15-23-E2	154.19

7	E3	7-14-13-9-8-16-24-25-E3	186.48
7	E5	7-14-13-9-10-18-20-28-33-31-E5	269.68
7	E6	7-14-13-9-8-16-24-25-27-32-E6	242.71
7	E7	7-14-13-9-10-18-20-28-33-E7	231.73
8	E4	8-9-13-21-29-E4	112.55
8	E5	8-16-24-25-27-32-33-31-E5	193.03
8	E6	8-16-24-25-27-32-E6	124.73
8	E7	8-16-24-25-27-32-33-E7	155.08
9	E3	9-8-16-24-25-E3	91.3
9	E4	9-13-21-29-E4	89.75
9	E5	9-10-18-20-28-33-31-E5	174.5
9	E6	9-8-16-24-25-27-32-E6	147.53
9	E7	9-10-18-20-28-33-E7	136.55
10	E1	10-9-8-16-24-E1	84.74
10	E3	10-9-8-16-24-25-E3	113.17
10	E4	10-9-13-21-29-E4	111.62
10	E5	10-18-20-28-33-31-E5	152.63
10	E6	10-18-20-28-27-32-E6	127.46

10	E7	10-18-20-28-33-E7	114.68
11	E1	11-9-8-16-24-E1	107.84
11	E3	11-9-8-16-24-25-E3	136.27
11	E4	11-14-22-30-29-E4	114.46
11	E5	11-9-10-18-20-28-33-31-E5	219.47
11	E6	11-9-8-16-24-25-27-32-E6	192.5
11	E7	11-9-10-18-20-28-33-E7	181.52
11	E8	11-14-22-30-34-E8	96.7
11	E9	11-14-22-30-34-E9	95.95
13	E3	13-9-8-16-24-25-E3	130.46
13	E5	13-9-10-18-20-28-33-31-E5	213.66
13	E6	13-9-8-16-24-25-27-32-E6	186.69
13	E7	13-9-10-18-20-28-33-E7	175.71
14	E1	14-13-9-8-16-24-E1	142.98
14	E3	14-13-9-8-16-24-25-E3	171.41
14	E5	14-13-9-10-18-20-28-33-31-E5	254.61
14	E6	14-13-9-8-16-24-25-27-32-E6	227.64
14	E7	14-13-9-10-18-20-28-33-E7	216.66

15	E4	15-8-9-13-21-29-E4	126.0
15	E5	15-23-E2-24-25-27-32-33-31-E5	192.78
15	E7	15-23-E2-24-25-27-32-33-E7	154.83
15	E8	15-8-9-13-14-22-30-34-E8	172.92
15	E9	15-8-9-13-14-22-30-34-E9	172.17
16	E4	16-8-9-13-21-29-E4	129.93
16	E5	16-24-25-27-32-33-31-E5	175.65
16	E6	16-24-25-27-32-E6	107.35
16	E7	16-24-25-27-32-33-E7	137.7
16	E8	16-8-9-13-14-22-30-34-E8	176.85
16	E9	16-8-9-13-14-22-30-34-E9	176.1
18	E1	18-10-9-8-16-24-E1	101.99
18	E2	18-10-9-8-15-23-E2	98.13
18	E3	18-10-9-8-16-24-25-E3	130.42
18	E4	18-10-9-13-21-29-E4	128.87
18	E5	18-20-28-33-31-E5	135.38
18	E6	18-20-28-27-32-E6	110.21
18	E7	18-20-28-33-E7	97.43

18	E8	18-10-9-13-14-22-30-34-E8	175.79
18	E9	18-10-9-13-14-22-30-34-E9	175.04
20	E1	20-28-27-25-24-E1	112.05
20	E2	20-28-27-25-24-E2	118.49
20	E3	20-28-27-25-E3	94.88
20	E4	20-18-10-9-13-21-29-E4	177.91
20	E6	20-28-27-32-E6	61.17
20	E8	20-18-10-9-13-14-22-30-34-E8	224.83
20	E9	20-18-10-9-13-14-22-30-34-E9	224.08
21	E1	21-13-9-8-16-24-E1	125.71
21	E2	21-13-9-8-15-23-E2	121.85
21	E3	21-13-9-8-16-24-25-E3	154.14
21	E5	21-13-9-10-18-20-28-33-31-E5	237.34
21	E6	21-13-9-8-16-24-25-27-32-E6	210.37
21	E7	21-13-9-10-18-20-28-33-E7	199.39
22	E1	22-14-13-9-8-16-24-E1	158.05
22	E2	22-14-13-9-8-15-23-E2	154.19
22	E3	22-14-13-9-8-16-24-25-E3	186.48

22	E5	22-14-13-9-10-18-20-28-33-31-E5	269.68
22	E6	22-14-13-9-8-16-24-25-27-32-E6	242.71
22	E7	22-14-13-9-10-18-20-28-33-E7	231.73
23	E4	23-15-8-9-13-21-29-E4	137.38
23	E5	23-E2-24-25-27-32-33-31-E5	181.4
23	E7	23-E2-24-25-27-32-33-E7	143.45
23	E8	23-15-8-9-13-14-22-30-34-E8	184.3
23	E9	23-15-8-9-13-14-22-30-34-E9	183.55
24	E4	24-16-8-9-13-21-29-E4	147.31
24	E5	24-25-27-32-33-31-E5	158.27
24	E7	24-25-27-32-33-E7	120.32
24	E8	24-16-8-9-13-14-22-30-34-E8	194.23
24	E9	24-16-8-9-13-14-22-30-34-E9	193.48
25	E4	25-24-16-8-9-13-21-29-E4	170.11
25	E5	25-27-32-33-31-E5	135.47
25	E7	25-27-32-33-E7	97.52
25	E8	25-24-16-8-9-13-14-22-30-34-E8	217.03
25	E9	25-24-16-8-9-13-14-22-30-34-E9	216.28

27	E4	27-25-24-16-8-9-13-21-29-E4	215.08
27	E8	27-25-24-16-8-9-13-14-22-30-34-E8	262.0
27	E9	27-25-24-16-8-9-13-14-22-30-34-E9	261.25
28	E1	28-27-25-24-E1	94.8
28	E2	28-27-25-24-E2	101.24
28	E3	28-27-25-E3	77.63
28	E4	28-20-18-10-9-13-21-29-E4	195.16
28	E8	28-20-18-10-9-13-14-22-30-34-E8	242.08
28	E9	28-20-18-10-9-13-14-22-30-34-E9	241.33
29	E1	29-21-13-9-8-16-24-E1	149.39
29	E2	29-21-13-9-8-15-23-E2	145.53
29	E3	29-21-13-9-8-16-24-25-E3	177.82
29	E5	29-21-13-9-10-18-20-28-33-31-E5	261.02
29	E6	29-21-13-9-8-16-24-25-27-32-E6	234.05
29	E7	29-21-13-9-10-18-20-28-33-E7	223.07
30	E1	30-22-14-13-9-8-16-24-E1	173.12
30	E2	30-22-14-13-9-8-15-23-E2	169.26
30	E3	30-22-14-13-9-8-16-24-25-E3	201.55

30	E5	30-22-14-13-9-10-18-20-28-33-31-E5	284.75
30	E6	30-22-14-13-9-8-16-24-25-27-32-E6	257.78
30	E7	30-22-14-13-9-10-18-20-28-33-E7	246.8
31	E1	31-33-32-27-25-24-E1	114.54
31	E2	31-33-32-27-25-24-E2	120.98
31	E3	31-33-32-27-25-E3	97.37
31	E4	31-33-28-20-18-10-9-13-21-29-E4	215.21
31	E8	31-33-28-20-18-10-9-13-14-22-30-34-E8	262.13
31	E9	31-33-28-20-18-10-9-13-14-22-30-34-E9	261.38
32	E4	32-27-25-24-16-8-9-13-21-29-E4	229.14
32	E8	32-27-25-24-16-8-9-13-14-22-30-34-E8	276.06
32	E9	32-27-25-24-16-8-9-13-14-22-30-34-E9	275.31
33	E1	33-32-27-25-24-E1	111.74
33	E2	33-32-27-25-24-E2	118.18
33	E3	33-32-27-25-E3	94.57
33	E4	33-28-20-18-10-9-13-21-29-E4	212.41
33	E8	33-28-20-18-10-9-13-14-22-30-34-E8	259.33
33	E9	33-28-20-18-10-9-13-14-22-30-34-E9	258.58

34	E1	34-30-22-14-13-9-8-16-24-E1	188.19
34	E2	34-30-22-14-13-9-8-15-23-E2	184.33
34	E3	34-30-22-14-13-9-8-16-24-25-E3	216.62
34	E5	34-30-22-14-13-9-10-18-20-28-33-31-E5	299.82
34	E6	34-30-22-14-13-9-8-16-24-25-27-32-E6	272.85
34	E7	34-30-22-14-13-9-10-18-20-28-33-E7	261.87

Table 4.2: Results of Dynamic Optimisation, Scenario A

4.3.2 Scenario B

This analysis investigates the conditions under which the set $V = \{11, 16, 22\}$ can be classified as a collection of real-time blocked nodes. The computing results for this situation are presented in [Table 4.3](#). The system provides the most efficient alternative route to many exits within the static evacuation plan, considering the presence of blocked nodes.

Node	Exit	Shortest Path	Distance(m)
1	E1	1-8-15-23-E2-24-E1	70.57
1	E3	1-8-9-17-25-E3	86.12
1	E4	1-8-9-17-21-29-E4	123.71
1	E6	1-8-9-17-19-27-32-E6	138.87

1	E8	1-8-9-17-21-29-30-34-E8	187.85
1	E9	1-8-9-17-21-29-30-34-E9	187.1
2	E1	2-9-17-25-24-E1	80.73
2	E6	2-9-17-19-27-32-E6	116.31
2	E8	2-9-17-21-29-30-34-E8	165.29
2	E9	2-9-17-21-29-30-34-E9	164.54
3	E8	3-10-18-17-21-29-30-34-E8	186.58
3	E9	3-10-18-17-21-29-30-34-E9	185.83
4	E1	4-2-9-17-25-24-E1	125.7
4	E2	4-2-1-8-15-23-E2	121.28
4	E3	4-2-9-17-25-E3	108.53
4	E4	4-2-9-17-21-29-E4	146.12
4	E5	4-5-12-20-28-33-31-E5	142.56
4	E6	4-5-12-20-19-27-32-E6	114.2
4	E7	4-5-12-20-28-33-E7	104.61
4	E8	4-2-9-17-21-29-30-34-E8	210.26
4	E9	4-2-9-17-21-29-30-34-E9	209.51
5	E1	5-12-20-28-33-31-26-25-24-E1	139.03

5	E2	5-4-2-1-8-15-23-E2	143.0
5	E3	5-12-20-28-33-31-26-25-E3	121.86
5	E4	5-12-20-19-17-21-29-E4	167.26
5	E6	5-12-20-19-27-32-E6	92.48
5	E8	5-12-20-19-17-21-29-30-34-E8	231.4
5	E9	5-12-20-19-17-21-29-30-34-E9	230.65
6	E1	6-2-9-17-25-24-E1	116.89
6	E6	6-2-9-17-19-27-32-E6	152.47
6	E8	6-13-21-29-30-34-E8	138.41
6	E9	6-13-21-29-30-34-E9	137.66
7	E1	7-6-2-9-17-25-24-E1	157.34
7	E3	7-6-2-9-17-25-E3	140.17
7	E4	7-14-13-21-29-E4	106.61
7	E5	7-4-5-12-20-28-33-31-E5	182.7
7	E6	7-4-5-12-20-19-27-32-E6	154.34
7	E7	7-4-5-12-20-28-33-E7	144.75
7	E8	7-14-13-21-29-30-34-E8	170.75
7	E9	7-14-13-21-29-30-34-E9	170.0

8	E1	8-15-23-E2-24-E1	53.27
8	E3	8-9-17-25-E3	68.82
8	E4	8-9-17-21-29-E4	106.41
8	E6	8-9-17-19-27-32-E6	121.57
8	E8	8-9-17-21-29-30-34-E8	170.55
8	E9	8-9-17-21-29-30-34-E9	169.8
9	E1	9-17-25-24-E1	63.19
9	E6	9-17-19-27-32-E6	98.77
9	E8	9-17-21-29-30-34-E8	147.75
9	E9	9-17-21-29-30-34-E9	147.0
10	E8	10-18-17-21-29-30-34-E8	169.33
10	E9	10-18-17-21-29-30-34-E9	168.58
12	E2	12-20-28-33-31-26-25-24-E2	128.22
12	E4	12-20-19-17-21-29-E4	150.01
12	E6	12-20-19-27-32-E6	75.23
12	E8	12-20-19-17-21-29-30-34-E8	214.15
12	E9	12-20-19-17-21-29-30-34-E9	213.4
13	E1	13-9-17-25-24-E1	102.35

13	E6	13-9-17-19-27-32-E6	137.93
13	E8	13-21-29-30-34-E8	114.73
13	E9	13-21-29-30-34-E9	113.98
14	E1	14-13-9-17-25-24-E1	143.3
14	E3	14-13-9-17-25-E3	126.13
14	E4	14-13-21-29-E4	91.54
14	E5	14-7-4-5-12-20-28-33-31-E5	197.77
14	E6	14-7-4-5-12-20-19-27-32-E6	169.41
14	E7	14-7-4-5-12-20-28-33-E7	159.82
14	E8	14-13-21-29-30-34-E8	155.68
14	E9	14-13-21-29-30-34-E9	154.93
15	E4	15-8-9-17-21-29-E4	119.86
15	E8	15-8-9-17-21-29-30-34-E8	184.0
15	E9	15-8-9-17-21-29-30-34-E9	183.25
17	E1	17-25-24-E1	45.65
17	E2	17-25-24-E2	52.09
17	E8	17-21-29-30-34-E8	130.21
17	E9	17-21-29-30-34-E9	129.46

18	E2	18-26-25-24-E2	73.67
18	E8	18-17-21-29-30-34-E8	152.08
18	E9	18-17-21-29-30-34-E9	151.33
19	E2	19-27-25-24-E2	93.58
19	E4	19-17-21-29-E4	111.04
19	E8	19-17-21-29-30-34-E8	175.18
19	E9	19-17-21-29-30-34-E9	174.43
20	E4	20-19-17-21-29-E4	132.76
20	E8	20-19-17-21-29-30-34-E8	196.9
20	E9	20-19-17-21-29-30-34-E9	196.15
21	E1	21-17-25-24-E1	84.81
21	E2	21-17-25-24-E2	91.25
21	E6	21-17-19-27-32-E6	120.39
21	E8	21-29-30-34-E8	91.05
21	E9	21-29-30-34-E9	90.3
23	E4	23-E2-24-25-17-21-29-E4	129.54
23	E8	23-E2-24-25-17-21-29-30-34-E8	193.68
23	E9	23-E2-24-25-17-21-29-30-34-E9	192.93

24	E4	24-25-17-21-29-E4	106.41
24	E8	24-25-17-21-29-30-34-E8	170.55
24	E9	24-25-17-21-29-30-34-E9	169.8
25	E8	25-17-21-29-30-34-E8	147.75
25	E9	25-17-21-29-30-34-E9	147.0
26	E8	26-18-17-21-29-30-34-E8	169.33
26	E9	26-18-17-21-29-30-34-E9	168.58
27	E4	27-19-17-21-29-E4	125.1
27	E8	27-19-17-21-29-30-34-E8	189.24
27	E9	27-19-17-21-29-30-34-E9	188.49
28	E4	28-33-31-26-18-17-21-29-E4	142.49
28	E8	28-33-31-26-18-17-21-29-30-34-E8	206.63
28	E9	28-33-31-26-18-17-21-29-30-34-E9	205.88
29	E1	29-21-17-25-24-E1	108.49
29	E2	29-21-17-25-24-E2	114.93
29	E6	29-21-17-19-27-32-E6	144.07
30	E1	30-29-21-17-25-24-E1	149.44
30	E2	30-29-21-17-25-24-E2	155.88

30	E3	30-29-21-17-25-E3	132.27
30	E5	30-29-21-17-18-26-31-E5	209.2
30	E6	30-29-21-17-19-27-32-E6	185.02
30	E7	30-29-21-17-18-26-31-33-E7	176.85
31	E8	31-26-18-17-21-29-30-34-E8	186.58
31	E9	31-26-18-17-21-29-30-34-E9	185.83
32	E4	32-27-19-17-21-29-E4	139.16
32	E8	32-27-19-17-21-29-30-34-E8	203.3
32	E9	32-27-19-17-21-29-30-34-E9	202.55
33	E8	33-31-26-18-17-21-29-30-34-E8	189.38
33	E9	33-31-26-18-17-21-29-30-34-E9	188.63
34	E1	34-30-29-21-17-25-24-E1	164.51
34	E2	34-30-29-21-17-25-24-E2	170.95
34	E3	34-30-29-21-17-25-E3	147.34
34	E5	34-30-29-21-17-18-26-31-E5	224.27
34	E6	34-30-29-21-17-19-27-32-E6	200.09
34	E7	34-30-29-21-17-18-26-31-33-E7	191.92

Table 4.3: Results of Dynamic Optimisation, Scenario B

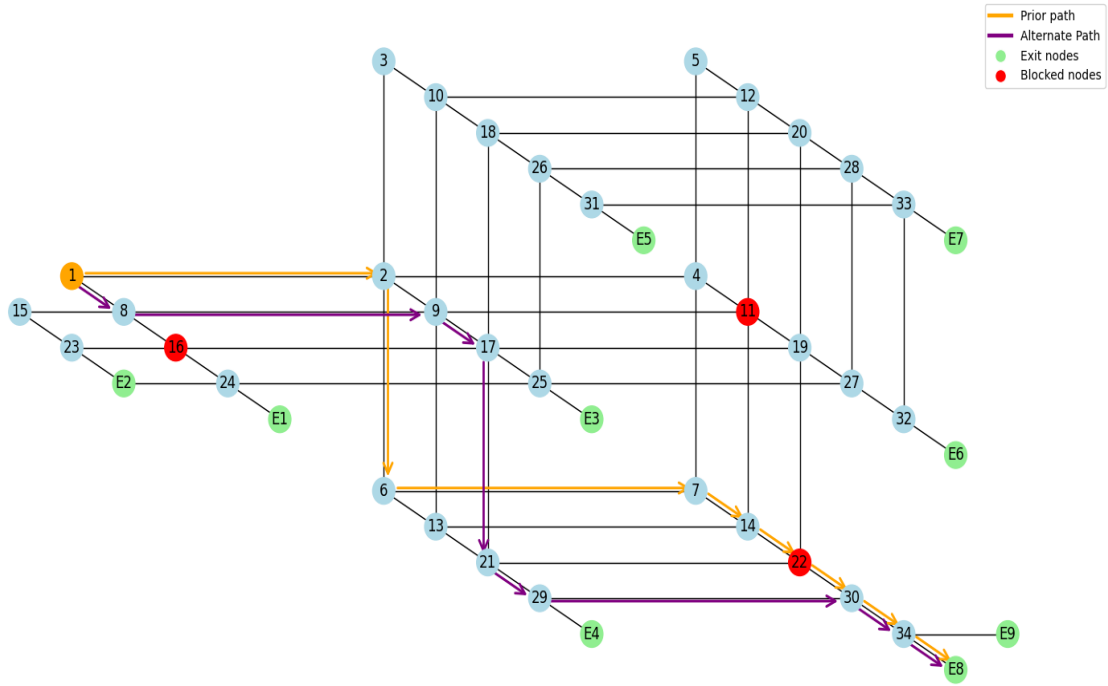


Figure 4.7: Prior and alternate shortest paths from node 1 to exit E8

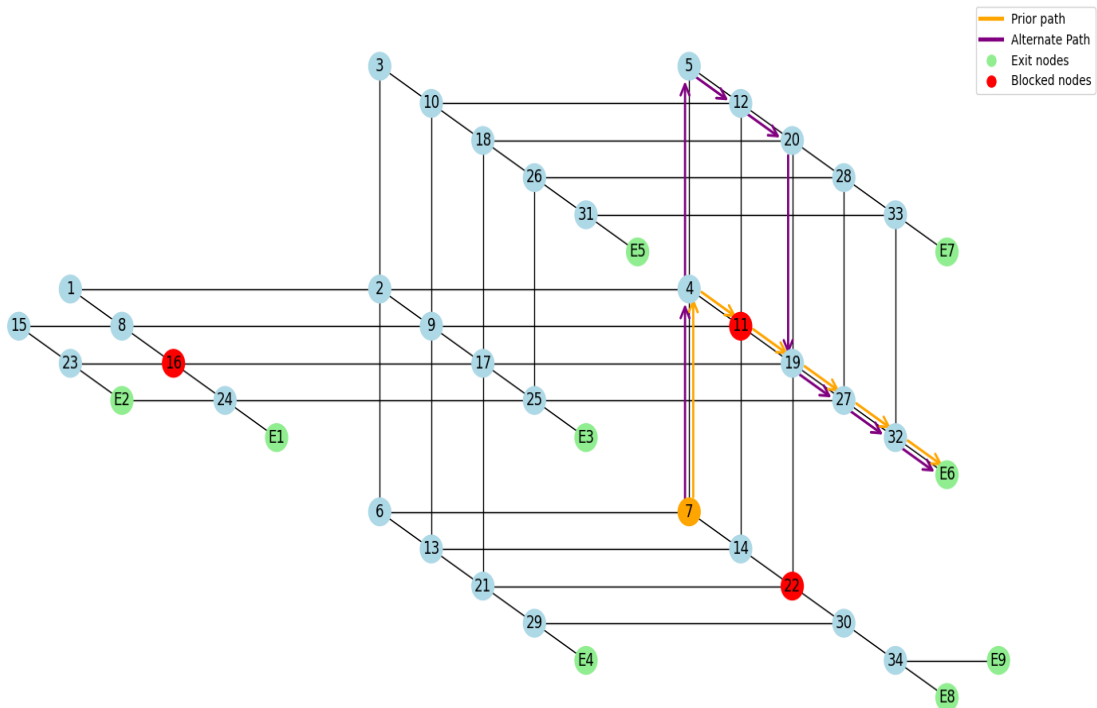


Figure 4.8: Prior and alternate shortest paths from node 7 to exit E6

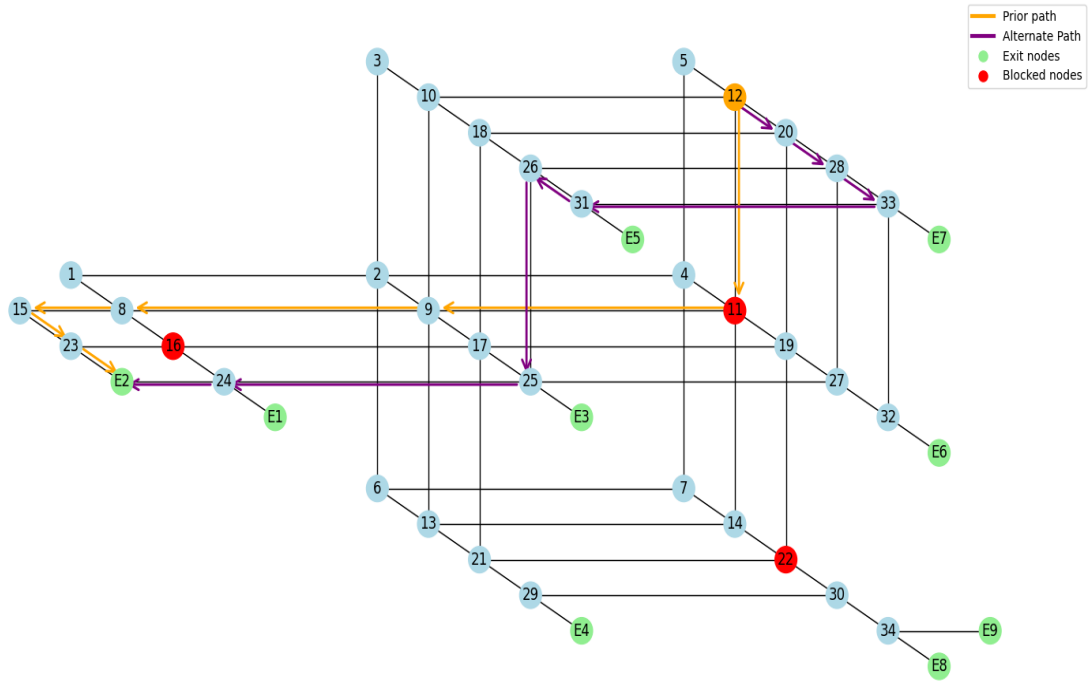


Figure 4.9: Prior and alternate shortest paths from node 12 to exit E2

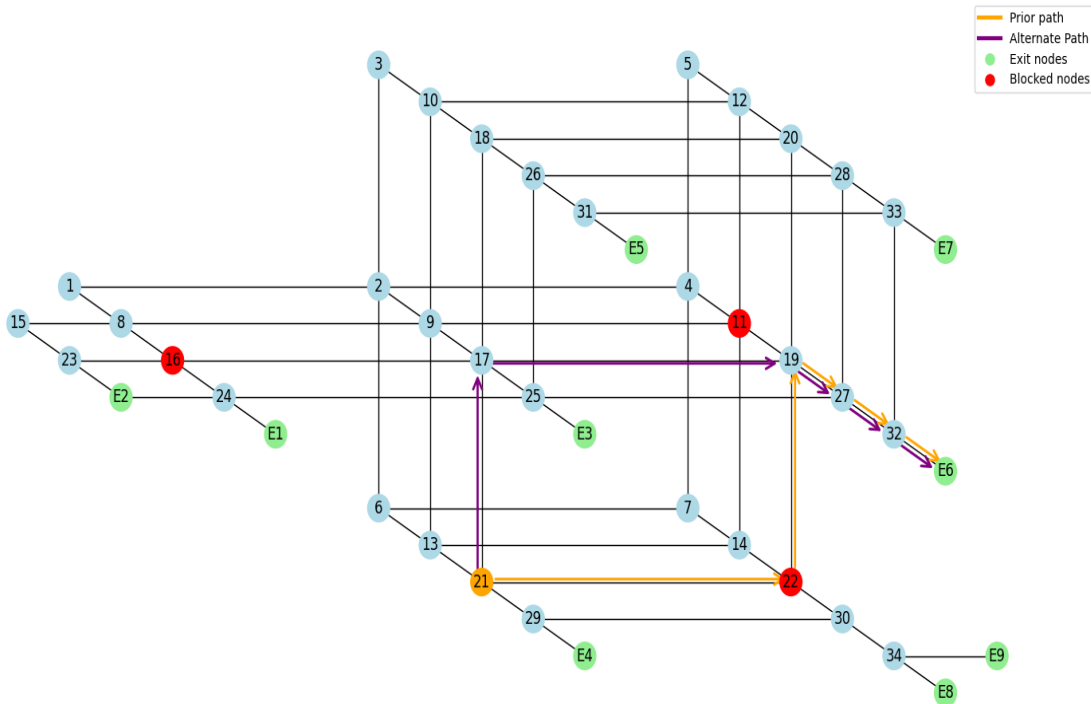


Figure 4.10: Prior and alternate shortest paths from node 21 to exit E6

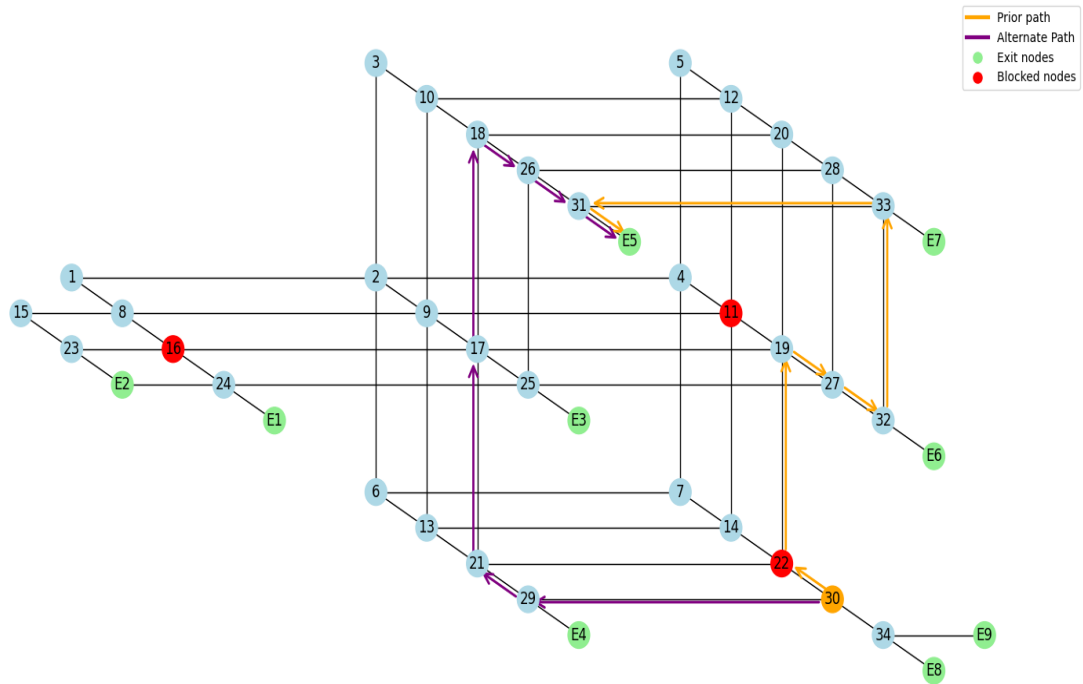


Figure 4.11: Prior and alternate shortest paths from node 30 to exit E5

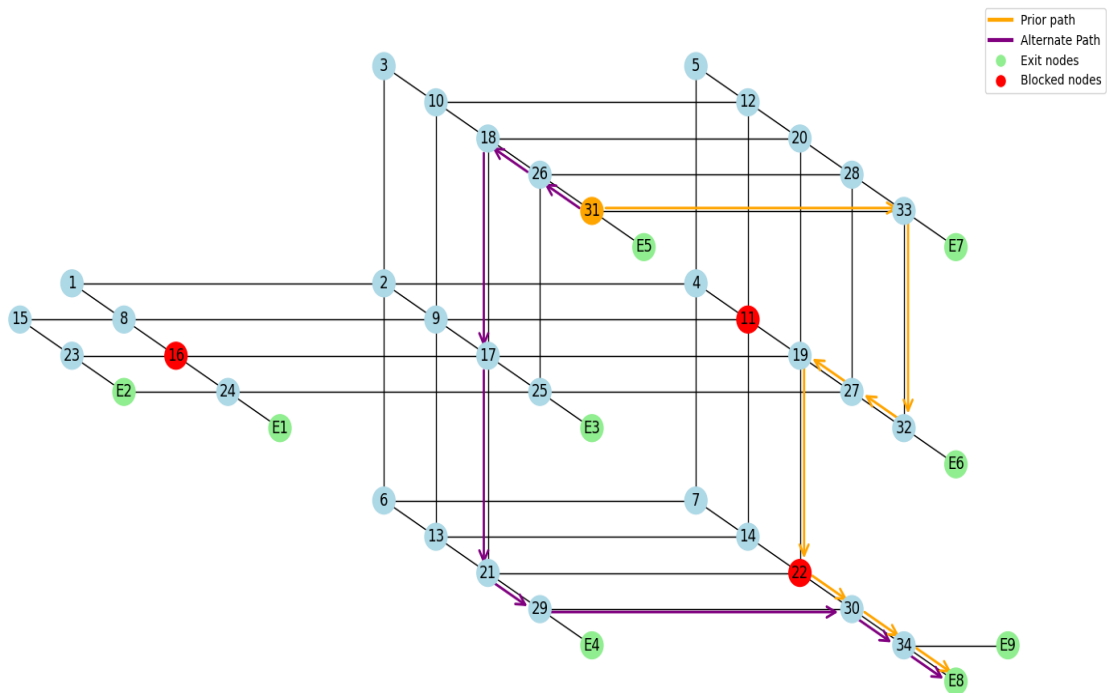


Figure 4.12: Prior and alternate shortest paths from node 31 to exit E8

The illustrations presented above depict graphical representations of scenario B, wherein nodes 11, 16, and 22 are obstructed. The presented information displays the most efficient routes to different exits and alternative routes that circumvent a certain obstructed node, starting from selected nodes. **Figure 4.7** shows the prior and alternate shortest path from node 1 to exit E8. **Figure 4.8**, **Figure 4.9** and **Figure 4.10** illustrate prior and alternate shortest paths from nodes 7, 12 and 21 to exits E6, E2 and E6 respectfully. The shortest paths from nodes 30 and 31 to exits E5 and E8 are shown in **Figure 4.11** and **Figure 4.12**.

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

In conclusion, a network model was constructed for the Aboagye Menyeh Complex, and optimisation models were developed for offline and real-time dynamic evacuations. Using the offline model, we successfully find the shortest path from any intersection within the Aboagye Menyeh Complex to any exit. Subsequently, various scenarios were conducted where some nodes were obstructed, and alternative routes to each exit were identified utilising the dynamic model.

5.2 Recommendation

The dynamic evacuation model developed in this study can be integrated into a sophisticated dynamic evacuation signage system. This system uses sensors and LED signage to dynamically alter the directions leading to exits, enabling individuals to circumvent hazardous areas within a building during emergencies. Other studies for various building complexes can be undertaken using different pathfinding methodologies.

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Chapter 6

Appendix A

Dijkstra Code

```
from heapq import heappush, heappop

def dijkstra(graph, source, destination):

    inf = float('inf')

    node_data = {}
    for node in graph:
        temp_dict = {"label": inf, "path": []}
        node_data[node] = temp_dict

    node_data[source]["label"] = 0

    temporary = []
    min_heap = []
    current_node = source

    while current_node != destination:
        if current_node not in temporary:
            temporary.append(current_node)
            for connected_node in graph[current_node]:
                if connected_node not in temporary:
                    new_label = node_data[current_node]["label"] + graph[current_node][connected_node]
                    if new_label < node_data[connected_node]["label"]:
                        node_data[connected_node]["label"] = new_label
                        node_data[connected_node]["path"] = node_data[current_node]["path"].copy()
                        node_data[connected_node]["path"].append(current_node)
                        heappush(min_heap, (node_data[connected_node]["label"], connected_node))

        current_path, current_node = heappop(min_heap)

    node_data[destination]["path"].append(destination)
    return node_data[destination]["path"]#, round(node_data[destination]["label"], 3)
```

Chapter 7

Appendix B

Graph Code

```
import networkx as nx
import matplotlib.pyplot as plt
from graph_data import ABComplex_data
from dikjstra import dikjstra

def shift_pos(pos, amount):
    new_pos = {}
    for node, (x, y) in pos.items():
        new_pos[node] = (x + amount, y+amount)
    return new_pos

def build_graph(source, destination):
    colours = ['violet', 'orange', 'yellow', 'indigo', 'blue', 'cyan', 'brown', 'violet']

    # Create graph
    G = nx.Graph()

    # Add Nodes
    nodes = [
        '1', '2', '3', '4', '5', '6', '7',
        '8', '9', '10', '11', '12', '13', '14',
        '15', '16', '17', '18', '19', '20', '21',
        '22', '23', '24', '25', '26', '27', '28',
        '29', '30', '31', '32', '33', '34',
        'E1', 'E2', 'E3', 'E4'
    ]
    G.add_nodes_from(nodes)

    # Extract and add edges
    edges = []
    for node in ABComplex_data:
        for neighbour_node in ABComplex_data[node]:
            if ((node, neighbour_node) in edges) or ((neighbour_node, node) in edges):
                continue

            if node.startswith("E"):
                if neighbour_node.startswith("E") and int(neighbour_node[1:]) > int(node[1:]):
                    edges.append((node, neighbour_node))
            else:
                if neighbour_node.startswith("E"):
                    edges.append((node, neighbour_node))
                else:
                    if int(neighbour_node) > int(node):
                        edges.append((node, neighbour_node))
```

```
G.add_edges_from(edges)
```

```
# Set positions
```

```
position = {
    '1': (0,0), '2': (3,0), '3': (3,3), '4': (6,0), '5': (6,3), '6': (3,-3), '7': (6,-3),
    '8': (0.5,-0.5), '9': (3.5,-0.5), '10': (3.5,2.5), '11': (6.5, -0.5),
    '12': (6.5,2.5), '13': (3.5,-3.5), '14': (6.5, -3.5), '15': (-0.5,-0.5),
    '16': (1,-1), '17': (4,-1), '18': (4,2), '19': (7,-1), '20': (7,2), '21': (4,-4),
    '22': (7,-4), '23': (0,-1), '24': (1.5, -1.5), '25': (4.5,-1.5), '26': (4.5, 1.5),
    '27': (7.5,-1.5), '28': (7.5, 1.5), '29': (4.5, -4.5), '30': (7.5,-4.5), 'E2': (0.5,-1.5),
    '31': (5,1), '32': (8,-2), '33': (8,1), '34': (8,-5), 'E1': (2,-2), 'E4': (5,-5), 'E3': (5,-2),
    'E5': (5.5, 0.5), 'E7': (8.5,0.5), 'E6': (8.5,-2.5), 'E8': (8.5,-5.5), 'E9': (9,-5),
}
```

```
# colour particular nodes
```

```
exit_nodes = ['E1', 'E2', 'E3', 'E4', 'E5', 'E6', 'E7', 'E8', 'E9']
node_colors = {}
```

```
for node in exit_nodes:
    node_colors[node] = 'lightgreen'
```

```
node_colors[source] = 'orange'
```

```
fig, ax = plt.subplots()
```

```
# Draw graph outline
```

```
nx.draw(G, position, with_labels=True,
        node_color=[node_colors.get(node, 'lightblue') for node in G.nodes()], node_size=400)
```

```
pos = [0.04, -0.04]
```

```
shortest_path = dijkstra(ABComplex_data, source, destination)
```

```
highlighted_edges = [(shortest_path[i], shortest_path[i+1]) for i in range(len(shortest_path)-1)]
```

```
nx.draw_networkx_edges(G, shift_pos(position, 0.04), edgelist=highlighted_edges, edge_color='orange',
                      width=2, ax=ax, arrows=True, arrowstyle='->', arrowsize=20)
```

```
red_patch = plt.Line2D([], [], color='violet', linewidth=3, label= 'Node'+source)
```

```
ax.legend(handles=[red_patch])
```

```
# plt.savefig("./exit_1.png")
```

```
plt.show()
```

Chapter 8

Appendix C

Graph Data

```
graph_data = {
    "1": {"2": 4, "3": 6, "4": 8},
    "2": {"1": 4, "3": 1, "5": 7},
    "3": {"1": 6, "2": 1, "4": 2, "5": 3, "6": 4},
    "4": {"1": 8, "3": 2, "6": 5},
    "5": {"2": 7, "3": 3, "6": 1, "7": 6},
    "6": {"3": 4, "4": 5, "5": 1, "7": 8},
    "7": {"5": 6, "6": 8}
}

ABComplex_data = {
    "1": {"8": 17.3, "2": 22.8},
    "2": {"1": 22.8, "3": 21.87, "4": 44.97, "6": 36.16, "9": 17.54},
    "3": {"2": 21.87, "10": 17.25},
    "4": {"2": 44.97, "5": 21.72, "7": 40.14, "11": 14.06},
    "5": {"4": 21.72, "12": 17.25},
    "6": {"2": 36.16, "7": 40.95, "13": 23.68},
    "7": {"4": 40.14, "6": 40.45, "14": 15.07},
    "8": {"1": 17.3, "9": 22.8, "15": 13.45, "16": 17.38},
    "9": {"2": 17.54, "8": 22.8, "10": 21.87, "11": 44.97, "13": 39.16, "17": 17.54},
    "10": {"3": 17.25, "9": 21.87, "12": 49.04, "18": 17.25},
    "11": {"4": 14.06, "9": 44.97, "12": 21.72, "14": 40.14, "19": 14.06},
    "12": {"5": 17.25, "10": 49.04, "11": 21.72, "20": 17.25},
    "13": {"6": 23.68, "9": 39.16, "14": 40.95, "21": 23.68},
    "14": {"7": 15.07, "11": 40.14, "13": 40.95, "22": 15.07},
    "15": {"8": 13.45, "23": 11.38},
    "16": {"8": 17.38, "17": 22.8, "23": 13.45, "24": 17.38},
    "17": {"9": 17.54, "16": 22.8, "18": 21.87, "19": 44.97, "21": 39.16, "25": 17.54},
    "18": {"10": 17.25, "17": 21.87, "20": 49.04, "26": 17.25},
    "19": {"11": 14.06, "17": 44.97, "20": 21.72, "22": 40.14, "27": 14.06},
    "20": {"12": 17.25, "18": 49.04, "19": 21.72, "28": 17.25},
    "21": {"13": 23.68, "17": 39.16, "22": 40.95, "29": 23.68},
    "22": {"14": 15.07, "19": 40.14, "21": 40.95, "30": 15.07},
    "23": {"15": 11.38, "16": 13.45, "E2": 11.38},
    "24": {"16": 17.38, "25": 22.8, "E1": 5.31, "E2": 11.75},
    "25": {"17": 17.54, "24": 22.8, "26": 21.87, "27": 44.97, "E3": 10.94},
    "26": {"18": 17.25, "25": 21.87, "28": 49.04, "31": 17.25},
    "27": {"19": 14.06, "25": 44.97, "28": 21.72, "32": 14.06},
    "28": {"20": 17.25, "26": 49.04, "27": 21.72, "33": 17.25},
    "29": {"21": 23.68, "30": 40.95, "E4": 3.23},
    "30": {"22": 15.07, "29": 40.95, "34": 15.07},
    "31": {"26": 17.25, "33": 2.8, "E5": 49.04},
    "32": {"27": 14.06, "33": 24.6, "E6": 8.14},
    "33": {"28": 17.25, "31": 2.8, "32": 24.6, "E7": 13.89},
    "34": {"30": 15.07, "E8": 11.35, "E9": 10.60},
    "E1": {"24": 5.31},
    "E2": {"23": 11.38, "24": 11.75},
```

```
"E3": { "25": 10.94 } ,  
"E4": { "29": 3.23 } ,  
"E5": { "31": 49.04 } ,  
"E6": { "32": 8.14 } ,  
"E7": { "33": 13.89 } ,  
"E8": { "34": 11.35 } ,  
"E9": { "34": 10.60 }  
}
```